

Faculty of Informatics Engineering  
Department of Software Engineering

## **Traffic Police Assistance System**

A senior 1 project report- Completed the requirements for obtaining a bachelor's degree in Informatics Engineering - Software Engineering and Information Systems

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كلية هندسة المعلوماتية  
قسم هندسة البرمجيات ونظم المعلومات

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## ABSTRACT

This project addresses the design and implementation of an **Intelligent Traffic Police Assistant System**, aimed at providing traffic officers with a smart and integrated technological platform to support their daily operational tasks with greater efficiency and accuracy. Classified within the domain of **intelligent information systems and artificial intelligence applications in traffic management**, the project contributes to improving traffic violation handling processes and enhancing overall road safety.

The motivation behind this project arises from the continued reliance on traditional and manual procedures for recording traffic violations, which often lead to operational delays, susceptibility to human error, and difficulties in managing and analyzing violation data. Furthermore, many existing systems lack effective integration with modern artificial intelligence technologies, such as optical character recognition and speech processing, limiting their usability in real-world, field-based environments.

The primary objective of the project is to develop a reliable and intelligent system that enables traffic police officers to efficiently record traffic violations through **automatic vehicle data extraction from images using Optical Character Recognition (OCR)**, as well as **voice-based reporting through Speech-to-Text (STT) technologies**. The system also facilitates the structured management of traffic records, violation logs, and the generation of analytical reports to support monitoring and decision-making processes.

The project follows an analytical and applied methodology, beginning with a comprehensive analysis of system and user requirements, followed by the design of a normalized relational database using MySQL. Backend services are implemented using the Laravel framework, adopting a **microservices-based and message-driven architecture** with RabbitMQ to ensure scalability and loose coupling. The client-side application is developed using Flutter to deliver a responsive and user-friendly interface suitable for mobile devices. Testing and validation were conducted using tools such as Postman to ensure the correctness and reliability of the system's APIs, while Firebase was optionally integrated to support real-time notification functionality.

The final outcome is a coherent, intelligent, and user-centered traffic police assistant system that enhances operational efficiency, reduces reporting time, improves data accuracy, and supports traffic officers in enforcing regulations more effectively, thereby contributing to safer and more organized road traffic management.

## الملخص

هذا النظام تصميم وتنفيذ نظام مساعد ذكي لشرطة المرور، يهدف إلى تزويد رجال المرور بمنصة تقنية ذكية ومتكاملة لدعم مهامهم التشغيلية اليومية بكفاءة ودقة أعلى. ويُصنّف المشروع ضمن مجال أنظمة المعلومات الذكية وتطبيقات الذكاء الاصطناعي في إدارة المرور، حيث يسهم في تحسين آليات التعامل مع المخالفات المرورية وتعزيز مستوى السلامة على الطرق.

تنطلق دوافع هذا المشروع من الاعتماد المستمر على الإجراءات التقليدية واليدوية في تسجيل المخالفات المرورية، والتي غالبًا ما تؤدي إلى تأخير العمليات، وزيادة احتمالية الأخطاء البشرية، وصعوبة إدارة وتحليل بيانات المخالفات. إضافة إلى ذلك، تفتقر العديد من الأنظمة الحالية إلى التكامل الفعّال مع تقنيات الذكاء الاصطناعي الحديثة، مثل التعرف الضوئي على الحروف (OCR) ومعالجة الكلام، مما يحد من فعاليتها في البيانات الميدانية الواقعية.

يهدف المشروع بشكل أساسي إلى تطوير نظام ذكي وموثوق يمكّن رجال المرور من تسجيل المخالفات بكفاءة، وذلك من خلال الاستخراج التلقائي لبيانات المركبات من الصور باستخدام تقنية OCR، إلى جانب إعداد التقارير الصوتية باستخدام تقنيات تحويل الكلام إلى نص (STT). كما يوفّر النظام إدارة منظمة لسجلات المرور، وسجلات المخالفات، وإنتاج تقارير تحليلية تدعم عمليات المراقبة واتخاذ القرار.

يعتمد المشروع على منهجية تحليلية وتطبيقية، تبدأ بتحليل شامل لمتطلبات النظام والمستخدمين، يليها تصميم قاعدة بيانات علائقية مُطبّعة باستخدام MySQL. تم تنفيذ خدمات الخلفية (Backend) باستخدام إطار العمل Laravel، مع اعتماد بنية معمارية قائمة على الخدمات المصغرة والتراسل بالرسائل (Message-Driven Architecture) باستخدام RabbitMQ لضمان القابلية للتوسع وتقليل الترابط بين المكونات. أما تطبيق الواجهة الأمامية فقد تم تطويره باستخدام Flutter لتوفير واجهة استخدام تفاعلية ومتجاوبة تناسب الأجهزة المحمولة. وقد أجريت عمليات الاختبار والتحقق باستخدام أدوات مثل Postman لضمان صحة وموثوقية واجهات برمجة التطبيقات (APIs)، مع دمج Firebase بشكل اختياري لدعم الإشعارات الفورية.

وتتمثل النتيجة النهائية في تقديم نظام مساعد ذكي ومتكامل يركز على المستخدم، يسهم في رفع الكفاءة التشغيلية، وتقليل زمن إعداد التقارير، وتحسين دقة البيانات، ودعم رجال المرور في تطبيق الأنظمة بفعالية أكبر، مما يساهم في تحقيق إدارة مرورية أكثر أمانًا وتنظيمًا.

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## List of abbreviations

| Abbreviation | Full Term                         |
|--------------|-----------------------------------|
| API          | Application Programming Interface |
| DB           | Database                          |
| DBMS         | Database Management System        |
| SQL          | Structured Query Language         |
| JWT          | JSON Web Token                    |
| CRUD         | Create, Read, Update, Delete      |
| OCR          | Optical Character Recognition     |
| STT          | Speech-to-Text                    |
| AI           | Artificial Intelligence           |
| UI           | User Interface                    |
| HTTP         | HyperText Transfer Protocol       |
| URL          | Uniform Resource Locator          |
| JSON         | JavaScript Object Notation        |
| OOP          | Object-Oriented Programming       |
| RMQ          | RabbitMQ Message Broker           |

# Chapter 1 Introduction



## Introduction:

In this chapter, we will introduce our project, discussing the main issues and reasons for building this system. We will also explain the objectives and goals we aim to accomplish with this system. Finally, we will provide an overview the report organization.

## Problem Definition:

In the field of traffic management and law enforcement, traffic police officers are responsible for maintaining road safety and enforcing regulations. However, many traffic monitoring and violation reporting processes still rely on manual methods, which are often inefficient, time-consuming, and prone to human error.

Traditional traffic violation systems commonly lack real-time processing, intelligent automation, and effective analytical capabilities, making it difficult to accurately manage and analyze large volumes of violation data. These limitations reduce operational efficiency and hinder effective field support for officers.

This project addresses these challenges by developing a **Traffic Police Assistant System** that enables officers to efficiently record traffic violations using **OCR** for automatic vehicle data extraction and **STT** for converting voice notes into structured text. The system also supports organized violation management and report generation.

The system is developed using modern technologies, including Laravel for backend services, Flutter for the mobile interface, and MySQL for database management, following a service-oriented and microservices-based architecture. Testing tools such as Postman are used to ensure reliability, with optional Firebase integration for notifications.

By modernizing traffic violation reporting, the proposed system improves efficiency, reduces errors, and enhances data accuracy, ultimately supporting effective traffic law enforcement and improved road safety.

## Project objectives:

The primary objective of this project is to develop an **Traffic Police Assistant System** that supports traffic police officers in efficiently recording, managing, and monitoring traffic violations, while ensuring that the system's functionalities align with real-world field operations to enhance accuracy, usability, and overall performance.

To achieve this objective, the system provides a set of intelligent features that enable officers to capture vehicle images and automatically extract relevant information using **Optical Character Recognition (OCR)** technologies, as well as record verbal observations that are converted into structured textual data

through **Speech-to-Text (STT)** mechanisms. The system also facilitates the organization and classification of traffic violations, enabling efficient record management and retrieval.

In addition, the system generates structured reports and analytical summaries that assist officers and administrators in reviewing violation data, identifying traffic patterns, and supporting informed decision-making. These analytical tools help improve situational awareness and provide insights into recurring violations and high-risk areas.

Ultimately, the project aims to simplify and modernize traffic violation management by providing a reliable, intelligent, and data-driven platform that enhances operational efficiency, reduces human error, and contributes to improved traffic law enforcement and road safety.

### **Report organization:**

The report will be organized in the following chapters:

- Chapter 1: introduction.
- Chapter 2: Basic concepts and literature review.
- Chapter 3: project management.
- Chapter 4: System analysis.
- Chapter 5: System Design
- Chapter 6: practical implementation.
- Chapter 7: Conclusion.

# **Chapter 2**

## **Basic concepts and Literature Review**

## **Basic concepts :**

### **1 . Intelligent Traffic Police Assistant System**

An intelligent traffic police assistant system is a smart software solution designed to support traffic officers in recording, managing, and analyzing traffic violations efficiently. The system combines automation and artificial intelligence to reduce manual work and enhance operational accuracy in field environments.

### **2. Optical Character Recognition (OCR)**

OCR is a core concept in the system, used to automatically extract vehicle license plate numbers from captured images. This reduces manual data entry, minimizes human error, and speeds up the traffic violation registration process.

### **3. Speech to Text (STT)**

Speech-to-Text technology enables traffic officers to report violations verbally. Spoken inputs are converted into structured textual data, allowing faster reporting and hands-free operation during traffic enforcement tasks.

### **4. mobile based field Operation :**

The system is primarily designed for mobile use, enabling traffic police officers to perform their duties directly from the field using smartphones or tablets. This concept ensures real-time data entry, accessibility, and operational flexibility.

### **5. Centralized Traffic Violation Database**

A centralized database stores all traffic-related records, including vehicle information, violation history, officer logs, and reports. Centralization ensures data consistency, secure access, and efficient data management.

### **6. Microservices-Based Backend Architecture**

The backend of the system is designed using a microservices architecture, where system functionalities are divided into independent services. This improves scalability, maintainability, and system reliability.

### **7. Message-Driven Communication (RabbitMQ)**

Message-driven communication enables asynchronous data exchange between system components. RabbitMQ is used to ensure reliable message delivery, reduce system coupling, and support high-volume traffic data processing.

## Literature Review :

### Similar Systems

#### 1. Traffic Police E-Ticketing Systems

Many countries have adopted **Electronic Traffic Ticketing (E-Ticketing) systems** that enable traffic officers to issue violations digitally using mobile devices. These systems allow officers to record vehicle details, violation types, and driver information directly into centralized databases, reducing paperwork and processing time. However, most e-ticketing systems rely heavily on **manual data entry** and lack advanced AI-based automation such as OCR or speech recognition.

#### 2. Automatic Number Plate Recognition (ANPR) Systems

ANPR systems are widely used in intelligent traffic surveillance and law enforcement. They utilize **computer vision and OCR techniques** to automatically detect and recognize vehicle license plates from images or video streams. These systems are effective for monitoring speeding, red-light violations, and stolen vehicles. Despite their accuracy, ANPR systems are often **standalone solutions** focused solely on plate recognition and do not provide integrated reporting, voice input, or officer-centered mobile workflows.

#### 3. Smart Traffic Management Systems

Smart traffic management platforms integrate traffic sensors, cameras, and centralized dashboards to monitor traffic flow and violations. These systems support decision-makers by providing analytics and statistical reports. However, they are primarily designed for **traffic control centers**, rather than direct use by traffic police officers in the field, and often lack mobile-friendly and voice-based interaction features.

#### 4. Mobile law Enforcement Reporting System

Some law enforcement agencies use mobile reporting systems that allow officers to file reports and logs using handheld devices. These systems improve mobility and reduce paperwork, but most of them do not integrate **OCR-based image processing** or **Speech-to-Text technologies**, limiting their effectiveness in traffic violation documentation.

## Comparison Between Existing Systems and the Proposed System :

| Feature                         | E-Ticketing Systems | ANPR Systems | Smart Traffic Systems | Proposed System |
|---------------------------------|---------------------|--------------|-----------------------|-----------------|
| Manual Data Entry               | Yes                 | No           | Partial               | <b>Yes</b>      |
| OCR Integration                 | No                  | Yes          | Yes                   | <b>Yes</b>      |
| Speech-to-Text (STT)            | No                  | No           | No                    | <b>Yes</b>      |
| Mobile Officer Support          | Yes                 | Limited      | No                    | <b>Yes</b>      |
| Integrated Violation Management | Partial             | No           | Partial               | <b>Yes</b>      |
| Analytical Reporting            | Limited             | Limited      | Yes                   | <b>Yes</b>      |
| AI-Based Automation             | Low                 | Medium       | Medium                | <b>High</b>     |

# **Chapter 3 Project Management**

## 1. Introduction:

In this chapter, we will dive into the management phase of the project, which is a critical aspect of ensuring the project's success. We will examine the project charter, project plan, Statement of Work (SOW) document, stakeholder analysis, and risk management strategies to effectively guide and control the project from initiation to completion.

## 2. Project charter:

A project charter is a formal document that serves as an official authorization for the start of a project. It acts as a reference point throughout the project, providing a clear understanding of the project's purpose and establishing a foundation for decision-making and project governance

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Project Title</b>      | Traffic police assistant                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Start Date</b>         | october 30, 2025                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Finish Date</b>        | June 10, 2026                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Project Manager(s)</b> | Dr. Riad Sonbol, Eng. Raghad Al Hosny                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Project Objectives</b> | <p>Automate violation recording using voice-to-text services, vehicle plate recognition, and automatically generate reports.</p> <p>Providing customized interfaces for each role (police_officer, police_manager, administrator, citizen).</p> <p>Enabling citizens to follow up on violations and submit appeals electronically</p> <p>Unifying data into a single platform that supports transparency and speed of decision-making</p> |
| <b>Project Approach</b>   | <p>The project will follow a step-by-step approach with testing after each phase:</p> <p>Determining the scope and objectives of the project.</p> <p>Performing a literature review of related systems and technologies.</p>                                                                                                                                                                                                              |



|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p>☐ Analyzing the functional and non-functional requirements of the system.</p> <p>Performing system decomposition and architectural design.</p> <p>Starting the first increment and implementing the authentication and account management services.</p> <p>Testing the functionalities of the first increment.</p> <p>Starting the second increment and implementing the traffic violation management services.</p> <p>Testing the functionalities of the second increment.</p> <p>Starting the third increment and implementing the artificial intelligence service for vehicle plate recognition (OCR).</p> <p>Testing the functionalities of the third increment.</p> <p>Starting the fourth increment and implementing the artificial intelligence service for speech-to-text processing.</p> <p>Testing the functionalities of the fourth increment.</p> <p>Documenting every step of the development process</p> |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Roles and Responsibilities:

|                         |                              |                                                                                |
|-------------------------|------------------------------|--------------------------------------------------------------------------------|
| Dr. Riad Sonbol         | Project Manager / Supervisor | General supervision and project oversight                                      |
| Raghad Al Hossny        | supervisor                   | Project coordination, progress monitoring, documentation support               |
| Mariam Albatool Almasri | Engineer                     | Backend development, API implementation , AI Contributor ,system documentation |
| Tasnim Aldakak          | Engineer                     | Frontend development, UI design, AI Contributor , system documentation         |

### 3. The SOW document

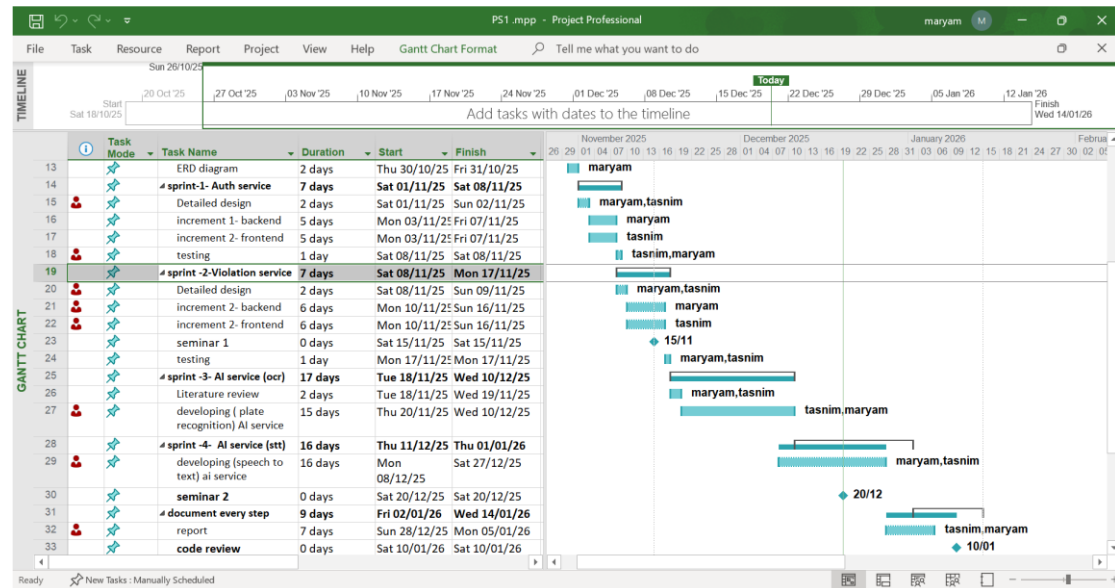
A Statement of Work (SOW) is a comprehensive document that defines the scope of work for a project. It outlines the specific tasks, deliverables, timeline, and responsibilities. The SOW document provides a clear understanding of what needs to be accomplished, the project's objectives, and the criteria for success.

|                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Project Title</b>       | Traffic police assistant                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Project Description</b> | An <b>intelligent system that assists traffic police officers</b> in recording and managing traffic violations using artificial intelligence technologies. The system supports <b>police officers</b> through a smart mobile application that facilitates violation documentation using AI-based services, provides <b>an administrator</b> responsible for managing user accounts and system settings, enables <b>citizens</b> to view their recorded violations and submit appeals electronically, and allows <b>police managers</b> to review violations and appeals and make approval or rejection decisions                                                                                                                                                                     |
| <b>Objectives</b>          | <p>Automate violation recording using voice-to-text services, vehicle plate recognition, and automatically generate reports.</p> <p>Providing customized interfaces for each role (police officer, police manager, administrator, citizen).</p> <p>Enabling citizens to follow up on violations and submit appeals electronically</p> <p>Unifying data into a single platform that supports transparency and speed of decision-making</p>                                                                                                                                                                                                                                                                                                                                            |
| <b>Project Scope</b>       | <p>Managing users and roles, including citizens, police officers, police managers, and administrators.</p> <p>Providing artificial intelligence services such as speech-to-text conversion, vehicle license plate recognition, and automatic violation report generation.</p> <p>Organizing and recording traffic violations by police officers through a smart field application.</p> <p>Enabling the Traffic Appeals Department to track appeal status from submission to final decision.</p> <p>Providing role-based control panels to display violations, statistics, and permitted operations.</p> <p>Maintaining a central database for storing violations, reports, and related attachments.</p> <p>Offering a citizen portal for tracking violations, submitting appeals</p> |
| <b>Project Goals</b>       | Automate traffic violation recording and documentation using artificial intelligence technologies.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                 | <p>Improve the accuracy and efficiency of traffic law enforcement operations.</p> <p>Enable electronic management and tracking of traffic violations and appeals.</p> <p>Provide role-based access and interactive dashboards for different system users.</p> <p>Enhance transparency by allowing citizens to view violations and submit appeals online.</p> <p>Support decision-making through centralized data storage and reporting</p> |
| <b>Deliverables</b>             | Project plan, SRS document, final report, backend and frontend system components, and artificial intelligence components including speech-to-text services, vehicle license plate recognition                                                                                                                                                                                                                                              |
| <b>Technologies Used</b>        | <p><b>Programming Languages &amp; Frameworks:</b><br/>Python, PHP, Django REST Framework, Laravel, Flutter</p> <p><b>Databases:</b><br/>MySQL, MongoDB</p> <p><b>Artificial Intelligence Technologies:</b><br/>Speech-to-Text models, Optical Character Recognition (OCR) models for vehicle plate recognition</p> <p><b>Tools:</b><br/>vs, GitHub, Postman, RabbitMQ</p>                                                                  |
| <b>Assumptions</b>              | <p>Team members are continuously available during development.</p> <p>Required AI technologies and development tools are accessible.</p> <p>The development environment remains stable throughout the project</p>                                                                                                                                                                                                                          |
| <b>Team Members &amp; Roles</b> | <p>Dr. Riad Sonbol – Project Manager</p> <p>Eng.raghad alhossny – supervisor</p> <p>Tasnim Aldakak – Frontend Developer, AI Contributor</p> <p>Mariam albatool Almasri – Backend Developer, AI Contributor</p>                                                                                                                                                                                                                             |
| <b>Schedule</b>                 | <p>Start: october 30, 2025</p> <p>First Seminar: november 15, 2025</p> <p>Second Project Seminar: December 20, 2025</p> <p>Technical Committee Review: January 11, 2026</p> <p>Finish: June 10, 2026</p> <p>Final Project Seminar: January 31, 2026</p>                                                                                                                                                                                    |

## 4. Project plan – Gantt chart:

A document outlining tasks, deadlines, and resources needed to achieve project objectives, serving as a roadmap for successful project execution.



## 5. Risk Management:

the process of identifying, evaluating, and mitigating potential risks that could impact the success of a project or a team.

| Risk_I<br>D | Risk title                | Risk Description                                                                                                               | Raised<br>Date | Trackin<br>g<br>Freque<br>ncy | State            | Impact | Mitigation Plan                                                                                           |
|-------------|---------------------------|--------------------------------------------------------------------------------------------------------------------------------|----------------|-------------------------------|------------------|--------|-----------------------------------------------------------------------------------------------------------|
| RK-01       | team of only two students | if one of the students stops for some reason, the whole project progress will be impacted                                      | 22/10/2025     | Weekly                        | active           | high   | the one who end first will help the other one, in his work                                                |
| RK-02       | Learning New Technologies | in our project, we need to use some AI components and our knowledge of developing it where low and advanced database concepts. | 12/11/2025     | Weekly                        | under mitigation | medium | Allocate time for learning, use official documentation, tutorials, and seek academic guidance when needed |

|              |                                                      |                                                                                                                                                                                           |            |        |        |        |                                                                                                                                                 |
|--------------|------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--------|--------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>RK-03</b> | Integration Challenges                               | Difficulties may arise when integrating backend APIs with the Flutter frontend, web frontend and AI services                                                                              | 30/10/2025 | Weekly | active | high   | Perform regular integration sprints, continuous API testing using Postman, and maintain clear API documentation                                 |
| <b>RK-04</b> | Data Security & Privacy                              | The system uses two databases (MySQL for core system data and MongoDB for AI-related data). Improper separation or security configuration may lead to data inconsistency or privacy risks | 20/10/2025 | Weekly | active | high   | Clearly separate database responsibilities, apply encryption and access control policies, and follow security best practices for both databases |
| RK-05        | R big Separation of work in the implementation stage | Dividing responsibilities between backend, frontend, and artificial intelligence components may lead to coordination challenges during the implementation stage                           | 15/12/2025 | daily  | active | medium | Assign an implementation leader and conduct regular coordination meetings to ensure alignment between all system components                     |

## 6. Version Control and Configuration Management:

### 6.1 Project Description

The *Traffic Police Assistance* project is a multi-service system developed using a microservices-based architecture. The system consists of backend services, frontend applications, and artificial intelligence services that work together to provide intelligent traffic violation management. Due to the complexity of the system and the involvement of multiple technologies and components, version control and configuration management were essential to ensure organized development, controlled integration, and system stability throughout the project lifecycle.

### 6.2 Git Repository Structure

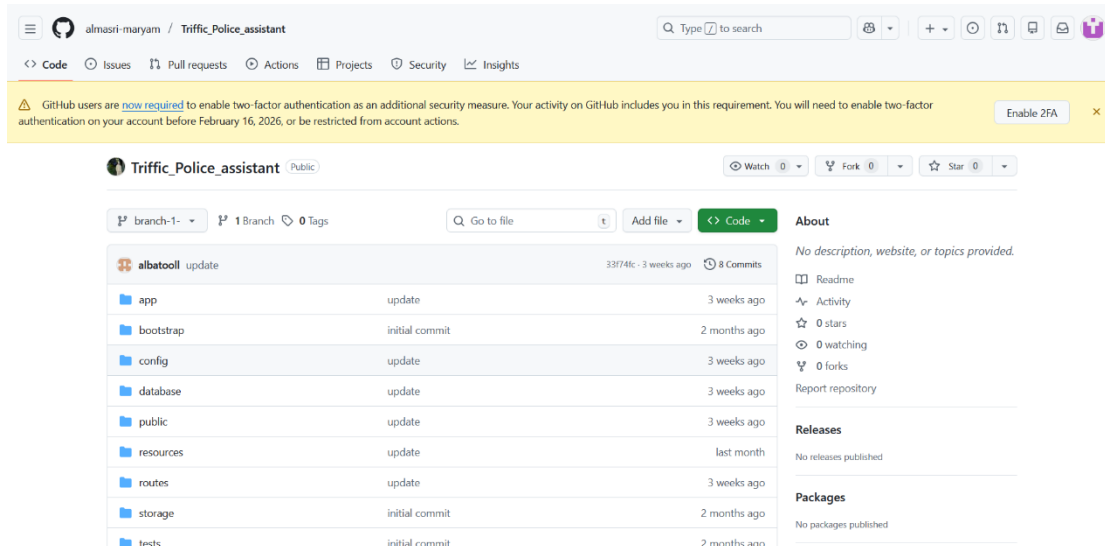
Version control for the project is managed using Git and hosted on GitHub. The backend source code is maintained in a dedicated repository that contains all backend microservices.

#### **Backend Repository:**

- **Repository Name:** `Triffic_police_assistant`
- **Purpose:** Contains the backend source code responsible for core system logic, user management, violation management, and integration with AI services.

The repository is structured to keep each microservice isolated, enabling independent development and deployment.

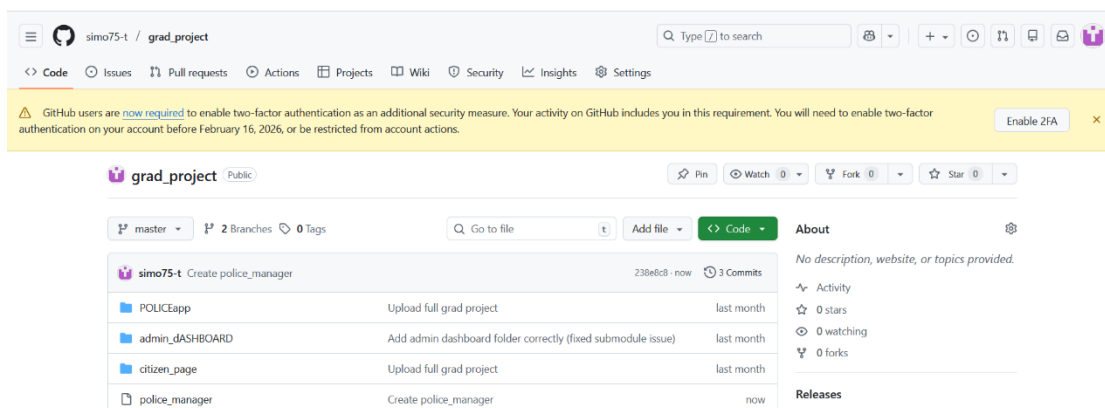
This separation improves maintainability and supports incremental system growth.



## Frontend Repository

**Purpose:** Contains the frontend source code for the web and mobile applications used by police officers, administrators, police managers, and citizens.

**Structure:** Organized into user interfaces, pages, services, and configuration files



## Frontend–Backend Integration

Although the frontend and backend are maintained in separate repositories, both follow a consistent version control strategy, including branch management, pull requests, and release tagging. This approach enables coordinated integration while preserving the ability to develop and release each part of the system independently.

## 6.3 Team Members and Responsibilities

The project was developed by a team of two members. All team members collaboratively participated in the system analysis and architectural design phases,

including requirements analysis, system modeling, and defining the overall system architecture.

#### **Backend Development:**

Responsible for implementing backend services using Django REST Framework and Laravel, including user management, traffic violation management, appeal handling, database design, and integration with artificial intelligence services

#### **Frontend Development:**

Responsible for managing frontend branches, implementing user interfaces, and integrating frontend logic with backend APIs.

#### **- AI Services Development:**

responsible for implementing artificial intelligence services, including speech-to-text processing, vehicle license plate recognition and communication with backend services

All team members contributed to version control activities. Backend- and frontend-related branches, merge operations, releases, and repository management were handled by the respective developers. Pull requests were used to merge feature branches into the development branch, ensuring controlled integration and traceability of changes.

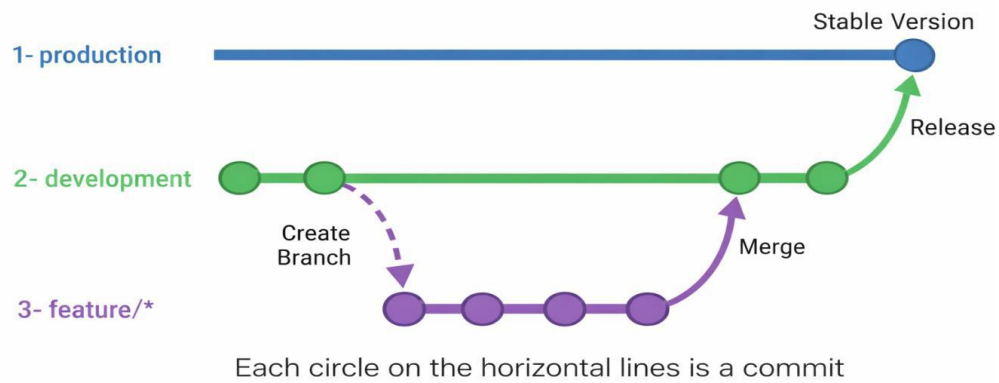
### **6.4 Development Workflow**

The development workflow for both the backend and frontend follows an incremental and task-based approach to ensure consistency, stability, and controlled integration across the entire system.

The workflow is summarized as follows:

1. A development task or feature is identified
2. A feature branch is created from the development branch
3. The feature is implemented and tested locally
4. A pull request is created to merge the feature branch into development
5. After integration and validation, the development branch is merged into production for release





## 7. Summary:

In conclusion, good project management is vital for successful software system development. It provides structure, ensuring projects are completed within scope and schedule. Overall, strong project management enhances the delivery of high quality software systems

# **Chapter 4**

## **System Analysis**

## Introduction:

In this chapter we will introduce the analytical study of the system using the needed UML diagrams for system requirements modeling.

## Requirements Elicitation:

| Req-ID   | Requirement Title                                                             | Description                                                                     | Category               | Actor          | Priority |
|----------|-------------------------------------------------------------------------------|---------------------------------------------------------------------------------|------------------------|----------------|----------|
| RE-FR-01 | The system shall allow police officers to log in                              | Requires entering a valid username and password to access the system            | Account Management     | Police Officer | High     |
| RE-FR-02 | The system shall allow police officers to log out                             | Enables the police officer to safely log out of the system                      | Account Management     | Police Officer | Medium   |
| RE-FR-03 | The system shall display and allow modification of police account information | Allows viewing and updating personal information such as phone number and email | Account Management     | Police Officer | Medium   |
| RE-FR-04 | The system shall allow recording a violation by voice                         | Enables creating a violation using voice commands converted into text           | Registering Violations | Police Officer | High     |
| RE-FR-05 | The system shall allow manual violation recording                             | Enables manual entry of violation details                                       | Registering Violations | Police Officer | High     |
| RE-FR-06 | The system shall allow capturing photos of violating vehicles                 | Enables taking photos as evidence of violations                                 | Registering Violations | Police Officer | High     |
| RE-FR-07 | The system shall allow review of AI-generated reports                         | Displays automated violation reports before approval                            | Report Management      | Police Officer | Medium   |
| RE-FR-08 | The system shall allow modification of AI-generated reports                   | Allows editing reports generated by artificial intelligence                     | Report Management      | Police Officer | Medium   |

|          |                                                                                   |                                                                 |                       |                |        |
|----------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------|-----------------------|----------------|--------|
| RE-FR-09 | The system shall display violation records created by the police officer          | Shows violations with status and history                        | Violations Management | Police Officer | High   |
| RE-FR-10 | The system shall allow searching for violations                                   | Enables searching violations using different criteria           | Violations Management | Police Officer | Medium |
| RE-FR-11 | The system shall allow citizens to view their violation records                   | Displays all violations recorded against the citizen            | Violations Management | Citizen        | High   |
| RE-FR-12 | The system shall allow electronic appeal submission                               | Enables citizens to submit and follow up appeals electronically | Appeals Management    | Citizen        | High   |
| RE-FR-13 | The system shall allow tracking of appeal status                                  | Displays appeal status (under review, accepted, rejected)       | Appeals Management    | Citizen        | High   |
| RE-FR-14 | The system shall allow police managers to log in                                  | Enables access to the administrative control panel              | Account Management    | Police Manager | High   |
| RE-FR-15 | The system shall allow police managers to log out                                 | Enables safe logout from the system                             | Account Management    | Police Manager | Medium |
| RE-FR-16 | The system shall allow police managers to view and edit their account information | Allows viewing and updating manager account data                | Account Management    | Police Manager | Medium |
| RE-FR-17 | The system shall allow administrators to create police accounts                   | Enables creation of new police officer accounts                 | Users Management      | Administrator  | High   |
| RE-FR-18 | The system shall allow administrators to                                          | Enables updating police officer account data                    | Users Management      | Administrator  | Medium |

|           |                                                                                 |                                                                    |                         |                |        |
|-----------|---------------------------------------------------------------------------------|--------------------------------------------------------------------|-------------------------|----------------|--------|
|           | modify police accounts                                                          |                                                                    |                         |                |        |
| RE-FR-19  | The system shall allow administrators to delete police accounts                 | Enables permanent deletion of police accounts                      | Users Management        | Administrator  | Medium |
| RE-FR-20  | The system shall allow administrators to activate or deactivate police accounts | Enables enabling or disabling police accounts                      | Users Management        | Administrator  | Medium |
| RE-FR-21  | The system shall allow administrators to view active police accounts            | Displays a list of all active police accounts                      | Users Management        | Administrator  | Medium |
| RE-FR-22  | The system shall allow police managers to review citizen appeals                | Enables reviewing appeals submitted by citizens                    | Appeals Management      | Police Manager | Medium |
| RE-FR-23  | The system shall convert voice commands into text                               | Converts spoken commands into processable text                     | Artificial Intelligence | AI             | High   |
| RE-FR-24  | The system shall analyze captured images                                        | Extracts vehicle plate number and type from images                 | Artificial Intelligence | AI             | High   |
| RE-FR-25  | The system shall generate automated violation reports                           | Generates reports including time, location, type, and plate number | Artificial Intelligence | AI             | High   |
| RE-FR-26  | The system shall classify violations                                            | Automatically classifies violations by type                        | Artificial Intelligence | AI             | High   |
| RE-NFR-01 | The system shall provide constant availability                                  | Ensures continuous service availability                            | Availability            | System         | High   |

|           |                                                 |                                                                |             |        |      |
|-----------|-------------------------------------------------|----------------------------------------------------------------|-------------|--------|------|
|           |                                                 | with minimal downtime                                          |             |        |      |
| RE-NFR-02 | The system shall support concurrent users       | Processes multiple requests simultaneously without degradation | Performance | System | High |
| RE-NFR-03 | The system shall be scalable                    | Allows the system to scale as usage increases                  | Scalability | System | High |
| RE-NFR-04 | The system shall secure data and communications | Encrypts all data and communications using HTTPS               | Security    | System | High |
| RE-NFR-05 | The system shall securely store passwords       | Uses strong hashing algorithms                                 | Security    | System | High |
| RE-NFR-06 | The system shall provide daily data backups     | Creates automated daily database backups                       | Reliability | System | High |

## SRS Document:

### Introduction:

#### Purpose:

The purpose of this document is to formally define the scope, architecture, and primary objectives of the Smart Traffic Police Assistant System. This system is designed to enhance the efficiency and accuracy of traffic violation management by leveraging artificial intelligence technologies, including speech-to-text processing and computer vision. The system supports traffic police officers in documenting violations through both manual and automated methods, facilitates administrative control and monitoring, and enables citizens to securely access their violation records and submit electronic appeals. Furthermore, the system aims to improve operational transparency, ensure data security, and provide a scalable and reliable digital solution that supports modern smart traffic management requirements.

#### Project scope:

The scope of the **Smart Traffic Police Assistant System** covers the design and implementation of a digital platform that supports traffic violation management through integrated artificial intelligence services. The system focuses on enabling traffic police officers to record violations using voice commands, manual data entry, and image capture, while automatically generating and classifying violation reports. It also includes administrative functionalities for managing police accounts, reviewing citizen appeals, and monitoring system operations. Additionally, the system provides citizens with secure access to their violation records and electronic appeal services. The scope of this project is limited to the software-based management of traffic violations and does not include physical traffic enforcement hardware or real-time traffic signal control.

#### Document overview:

The software requirements specification SRS document will be structured as follows:

1. Introduction.
2. Overall description and product features.
3. Nonfunctional requirements.
4. Conclusion.

## Overall description and product features:

### Product features:

The product features derived from the system's use cases include:

#### User Authentication (UC-01):

Allows police officers, police managers, administrators, and citizens to log in and log out of the system securely using authenticated credentials.

#### Manage Police Profile (UC-02):

Enables police officers to view and edit their profile information, such as personal and contact details.

#### Manage Police Accounts (UC-03):

Allows administrators to create, edit, activate, or deactivate police officer accounts within the system.

#### Record Violation Manually (UC-04):

Enables police officers to manually add a new traffic violation by entering violation details through the system.

#### Record and Transcribe Voice Commands (UC-05):

Allows police officers to record violation details using voice commands, which are automatically converted into text.

#### Capture and Analyze Vehicle Images (UC-06):

Enables capturing images of violating vehicles and extracting license plate information using image processing techniques.

#### Generate Violation Report (UC-07):

Automatically generates a structured violation report by combining manual input, voice transcription, and image analysis.

#### Manage Violations (UC-08):

Allows police officers to view, manage, and track recorded violations and their statuses.

#### Review Violations (UC-09):

Enables police managers to review recorded violations for monitoring and administrative purposes.

#### Submit Appeal (UC-10):

Allows citizens to submit an electronic appeal against a traffic violation.

#### Review Appeals (UC-11):

Enables police managers to review, accept, or reject appeals submitted by citizens.

#### View Violations for Citizens (UC-12):

Allows citizens to view their personal traffic violation records through the system



## **user class and charactrtistics**

The system supports the following user classes:

### **Police Officer:**

Primary users responsible for recording traffic violations, capturing evidence, generating violation reports, and managing violation records. They require moderate technical skills and use the system mainly in field operations.

### **Citizen:**

End users who access the system to view their traffic violations and submit appeals. They require minimal technical knowledge.

### **Police Manager:**

Supervisory users responsible for reviewing violations and handling citizen appeals. They require administrative and decision-making capabilities.

### **System Administrator:**

Privileged users responsible for managing police accounts, including creating, editing, activating, and deactivating accounts. This role does not interact directly with violation data.

## **Assumption**

### **Internet Connectivity:**

It is assumed that users will have access to a stable internet connection, as the system operates as an online platform.

### **Mobile Usage:**

The system is designed primarily for mobile use, especially for police officers working in the field.

### **Authorized System Access:**

All users are assumed to access the system using valid credentials assigned according to their roles.

### **Accuracy of Data Entry:**

Police officers are expected to enter accurate and complete violation information to ensure reliable reports and system outputs.

### **AI Assistance Limitation:**

Outputs generated by artificial intelligence components are considered supportive and subject to human review before final approval.

### Non functional requirements:

The non-functional requirements for the Smart Traffic Police Assistant System are as follows:

#### Usability:

The system shall provide user-friendly interfaces that enable traffic police officers, administrators, and citizens to perform core tasks efficiently. This includes simple and intuitive workflows for recording violations, reviewing reports, and submitting appeals, as well as clear visual cues such as meaningful icons and guided input forms to reduce user errors during field operations.

#### Security:

The system shall protect sensitive user data, including account information, violation records, and appeal details, by implementing secure authentication mechanisms, role-based access control, and encrypted communication channels (HTTPS). Additionally, an API Gateway shall be utilized to enforce access control policies and isolate backend services and databases from unauthorized access.

### Conclusion:

This software Requirements Specification (SRS) document outlines the critical functionalities, user roles, and assumptions essential for developing a robust and user-friendly system. By adhering to these requirements, the system will effectively support software product owners in enhancing their products to meet user needs and streamline project management processes.

## Requirements modeling:

### Use case diagram :

use-case diagrams model the behavior of a system and help to capture the requirements of the system.

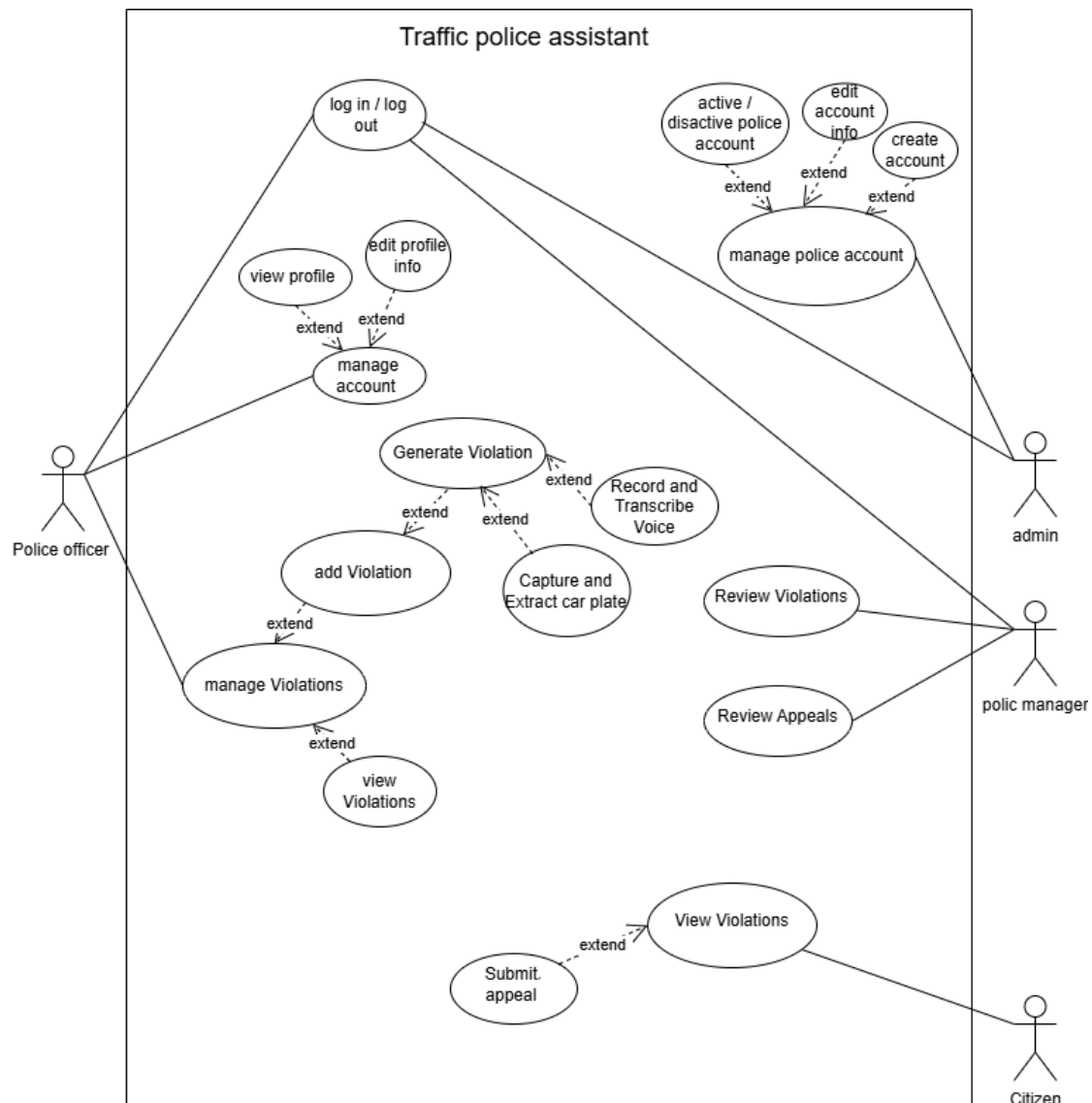


Figure 1 Use case diagram

## Uc -1-

### use case specification:

|                   |                                                                                                                                                                                                                                                                                                                                                     |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use case number   | Uc_1                                                                                                                                                                                                                                                                                                                                                |
| Title             | Login / Logout                                                                                                                                                                                                                                                                                                                                      |
| Description       | This feature allows police officers and administrators to securely log in and out of the system.                                                                                                                                                                                                                                                    |
| Actors            | Police Officer, Administrator                                                                                                                                                                                                                                                                                                                       |
| Precondition      | The user must have an active account.                                                                                                                                                                                                                                                                                                               |
| Main Flow         | <ol style="list-style-type: none"><li>1. The officer selects 'Login'.</li><li>2. The system displays the login form.</li><li>3. The user enters the username and password.</li><li>4. The system verifies the credentials.</li><li>5. If valid, access is granted.</li><li>6. To log out, the user selects 'Logout' and the session ends.</li></ol> |
| Alternative flows | First alternative flow A1:<br>Start at step 4 in the main flow where the credentials are invalid.<br>5. The system displays an error message and returns to step 2.                                                                                                                                                                                 |
| Postcondition     | The user session is successfully created or terminated.                                                                                                                                                                                                                                                                                             |

### sequence diagram

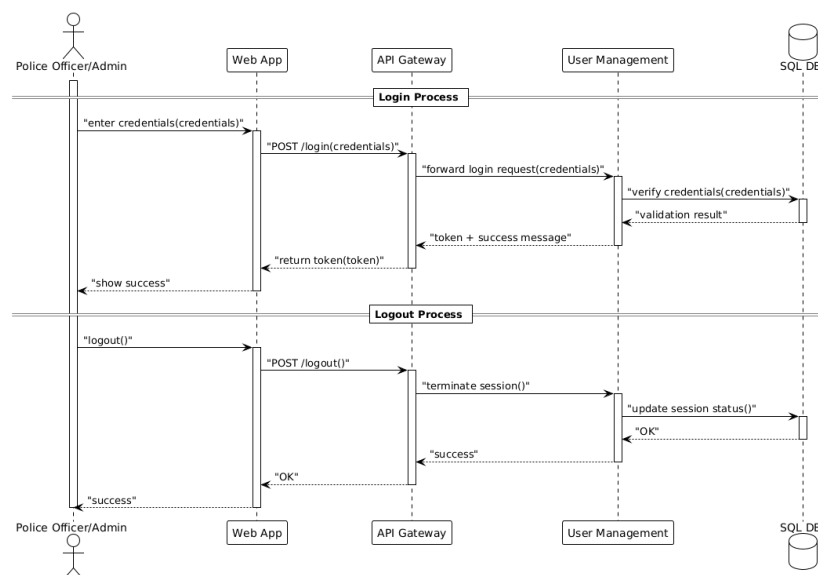


Figure 2 sequence diagram “Login / Logout”

## Uc -2-

### use case specification:

|                          |                                                                                                                                                                                                                                                                                                                            |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use case number</b>   | Uc_2                                                                                                                                                                                                                                                                                                                       |
| <b>Title</b>             | Account Management                                                                                                                                                                                                                                                                                                         |
| <b>Description</b>       | This feature allows the police officer to manage and update personal information.                                                                                                                                                                                                                                          |
| <b>Actors</b>            | Police Officer                                                                                                                                                                                                                                                                                                             |
| <b>Precondition</b>      | The officer must be logged in.                                                                                                                                                                                                                                                                                             |
| <b>Main Flow</b>         | 1. The officer selects 'View Profile'.<br>2. The system displays the account details.<br>3. if the officer need selects 'Edit Information'.<br>4. The system displays the edit form.<br>5. The officer enters the new data.<br>6. The system validates and saves the updates.<br>7. The system displays a success message. |
| <b>Alternative flows</b> | First alternative flow A1:<br>Start at step 6 where invalid data is entered.<br>7. The system displays an error message and returns to step 4.                                                                                                                                                                             |
| <b>Postcondition</b>     | The officer's profile information is updated successfully or the user is informed of any failure                                                                                                                                                                                                                           |

### Activity diagram:

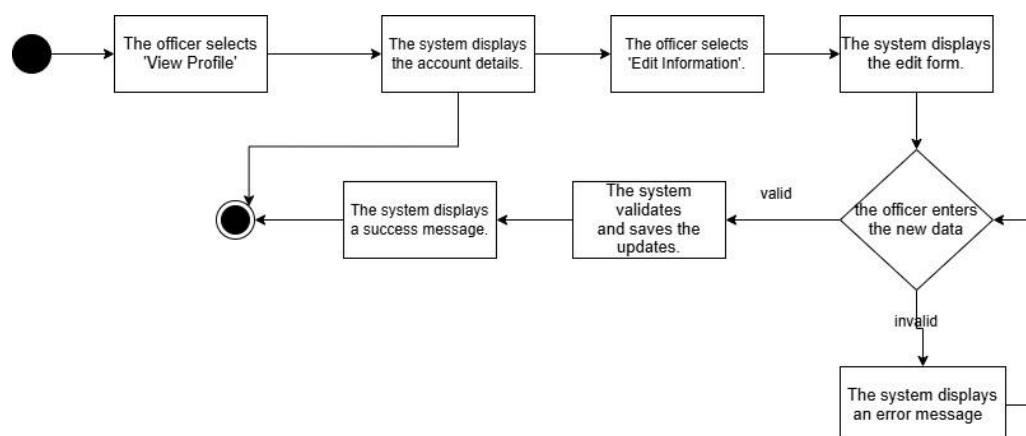


Figure 3 activity diagram “Account Management”

## sequence diagram

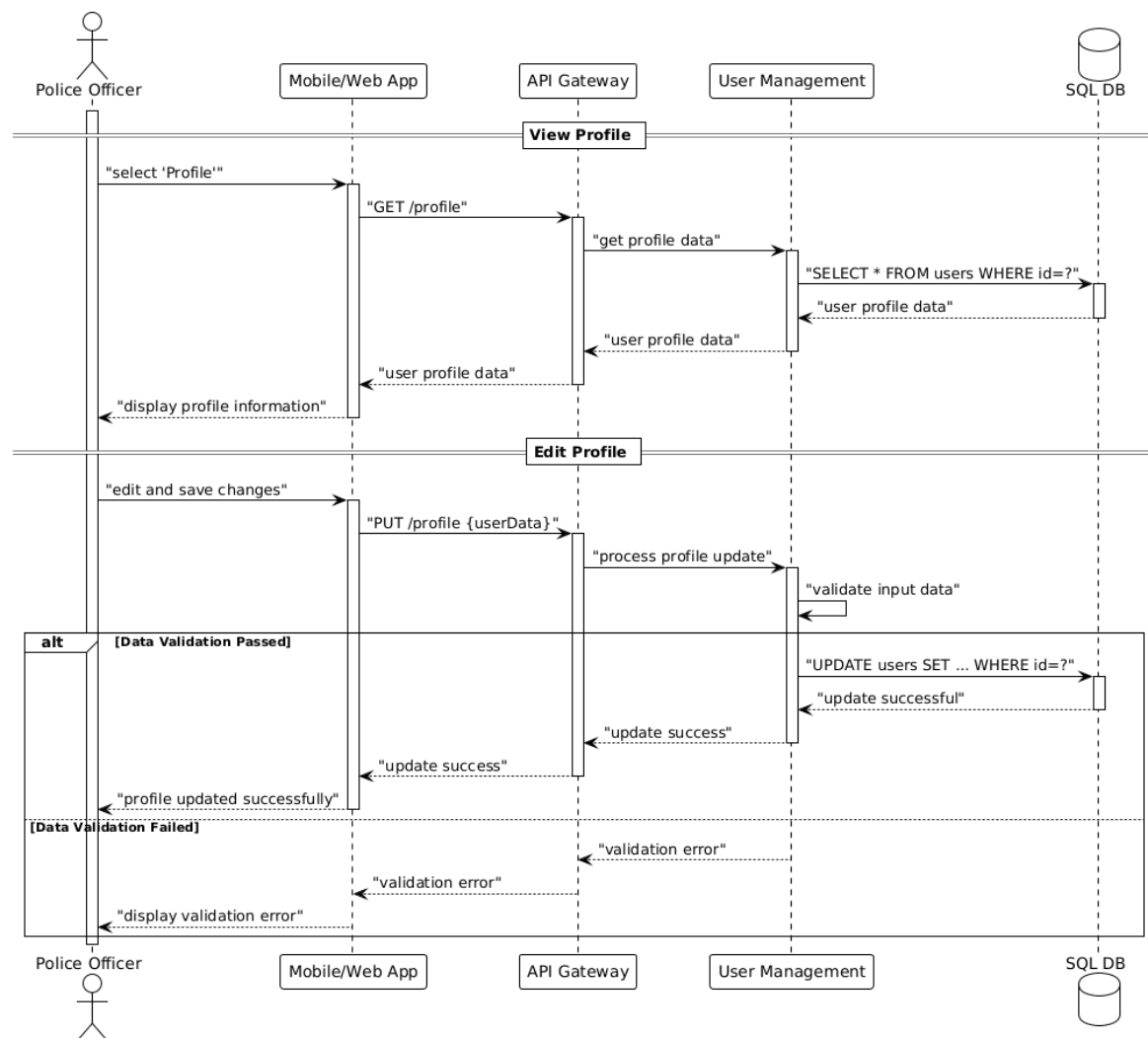


Figure 4 sequence diagram "Account Management"

### Uc -3-

#### use case specification:

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>use case number</b>  | UC_3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>title</b>            | Generate Violation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>description</b>      | This feature allows the police officer to create an AI-assisted traffic violation record.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>actors</b>           | Police Officer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>precondition</b>     | The officer must be logged in.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>main flow</b>        | <p>The officer selects 'Generate Violation'.</p> <p>2. The system prompts to record audio.</p> <p>3. The officer records the audio.</p> <p>4. The system converts it to text and displays it.</p> <p>5. The officer captures an image of the vehicle.</p> <p>6. The system extracts the license plate number.</p> <p>7. The officer selects the violation type.</p> <p>8. The system generates the violation.</p> <p>9.the officer review the violation</p> <p>10. the officer can edit the generated violation</p> <p>11. The officer confirms</p> <p>12. the system save the generated violation.</p> |
| <b>postcondition</b>    | The violation is successfully created or temporarily saved for later submission.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>alternative flow</b> | <p>First alternative flow A1:<br/>Start at step 3 where the recording fails.</p> <p>4. The system requests re-recording.</p> <p>Second alternative flow A2:<br/>Start at step 4 where conversion fails.</p> <p>5. The system allows manual text entry.</p> <p>Third alternative flow A3:<br/>Start at step 6 where extraction fails.</p> <p>7. The system asks to enter the plate number manually.</p>                                                                                                                                                                                                  |

## Activity diagram:

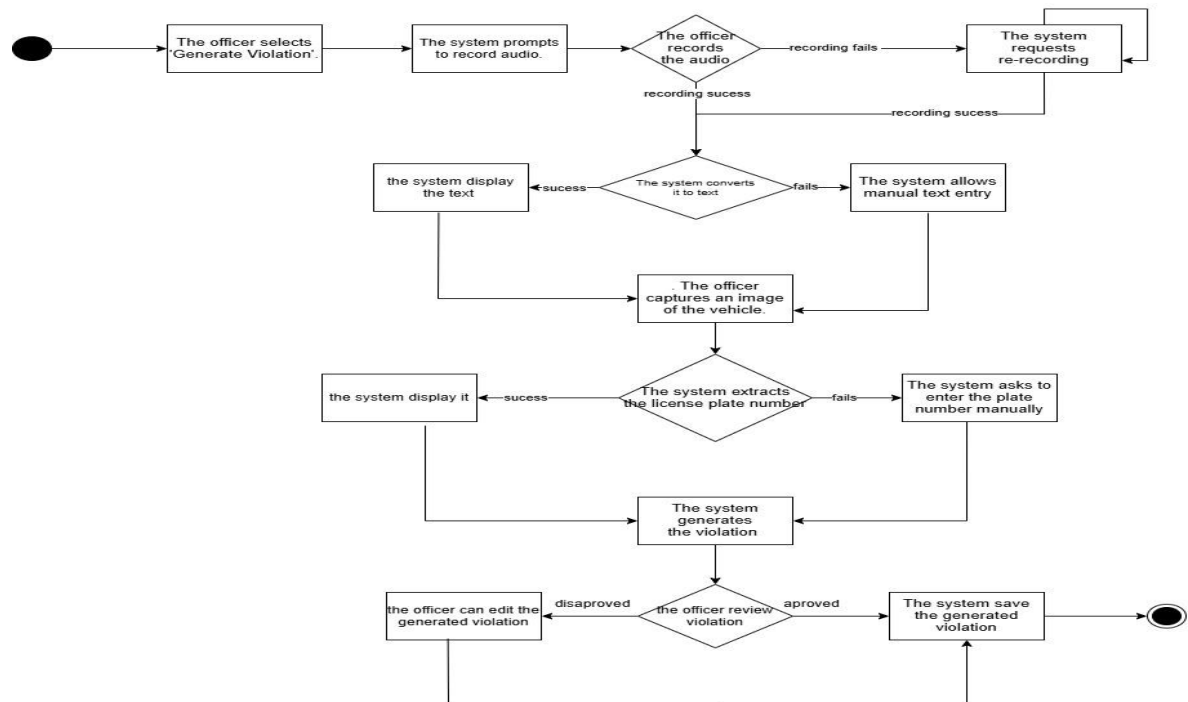


Figure 5 activity diagram “Generate Violation”

## sequence diagram

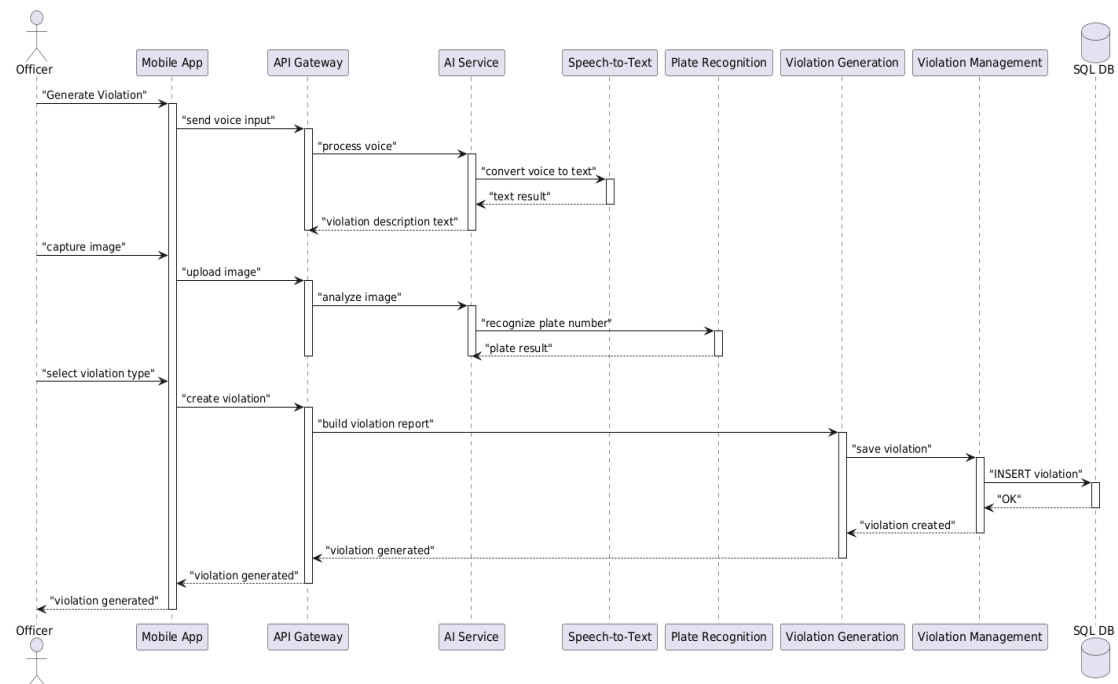


Figure 6 sequence diagram “Generate Violation”



#### Uc -4-

#### use case specification:

|                         |                                                                                                                                                                                                                                                                             |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use case number</b>  | UC_4                                                                                                                                                                                                                                                                        |
| <b>title</b>            | Add Violation                                                                                                                                                                                                                                                               |
| <b>description</b>      | This feature allows the police officer to manually add a new traffic violation.                                                                                                                                                                                             |
| <b>actors</b>           | Police Officer                                                                                                                                                                                                                                                              |
| <b>precondition</b>     | The officer must be logged in.                                                                                                                                                                                                                                              |
| <b>main flow</b>        | 1. The officer selects 'Add Violation'.<br>2. The system displays a data entry form.<br>3. The officer enters violation details.<br>4. The system validates the data.<br>5. The system generates and displays the violation report.<br>6. The officer reviews and confirms. |
| <b>postcondition</b>    | The violation is successfully saved.                                                                                                                                                                                                                                        |
| <b>alternative flow</b> | First alternative flow A1:<br>Start at step 4 where data is invalid or missing.<br>5. The system displays an error and returns to step 3.                                                                                                                                   |

#### Activity diagram:

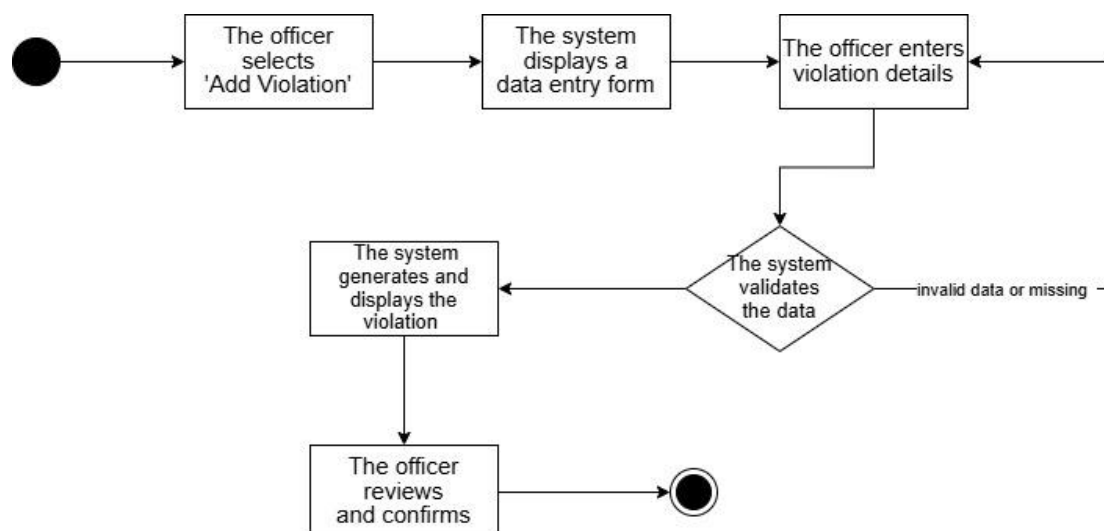


Figure 7 activity diagram “Add Violation”

## Uc -5-

### use case specification:

|                         |                                                                                                                                                                                           |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use case number</b>  | UC_5                                                                                                                                                                                      |
| <b>title</b>            | View Violation                                                                                                                                                                            |
| <b>description</b>      | This feature allows the police officer to view and manage existing violations.                                                                                                            |
| <b>actors</b>           | Police Officer                                                                                                                                                                            |
| <b>precondition</b>     | The officer must be logged in and have previous violations.                                                                                                                               |
| <b>main flow</b>        | 1. The officer selects 'View Violations'.<br>2. The system displays the list of violations.<br>3. If the officer need the can search by date or plate.<br>4. The system displays details. |
| <b>alternative flow</b> | First alternative flow A1:<br>Start at step 2 where no violations exist.<br>3. The system displays 'No violations found'.                                                                 |
| <b>postcondition</b>    | The violation details are successfully displayed.                                                                                                                                         |

### Activity diagram:

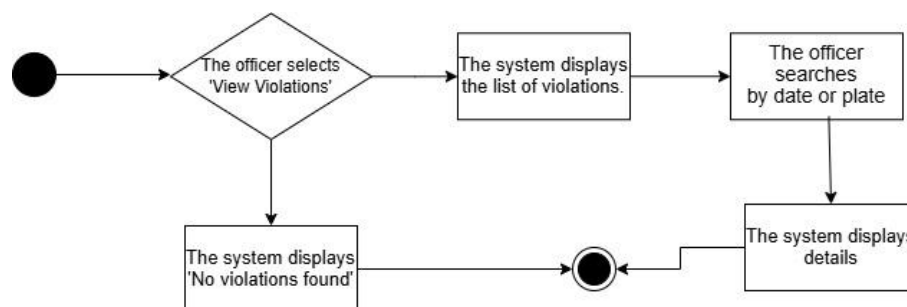


Figure 8 activity diagram “View Violation”

### sequence diagram:

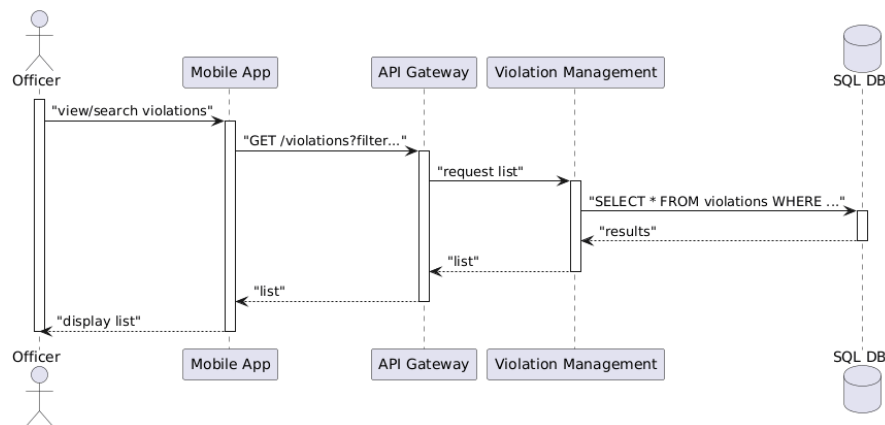


Figure 9 sequence diagram “View Violation

### Uc -6-

#### use case specification:

|                         |                                                                                                                                                                                                                                                                                                            |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use case number</b>  | UC_6                                                                                                                                                                                                                                                                                                       |
| <b>title</b>            | Create Police Account                                                                                                                                                                                                                                                                                      |
| <b>description</b>      | This feature allows the administrator to create new police officer accounts.                                                                                                                                                                                                                               |
| <b>actors</b>           | Administrator                                                                                                                                                                                                                                                                                              |
| <b>precondition</b>     | The administrator must be logged in.                                                                                                                                                                                                                                                                       |
| <b>main flow</b>        | <ol style="list-style-type: none"><li>1. The administrator selects 'Create New Account'.</li><li>2. The system displays the registration form.</li><li>3. The administrator enters officer details.</li><li>4. The system validates and saves the data.</li><li>5. The system confirms creation.</li></ol> |
| <b>postcondition</b>    | The new police account is successfully created.                                                                                                                                                                                                                                                            |
| <b>Alternative flow</b> | First alternative flow A1:<br>Start at step 4 where data is incomplete or duplicate.<br>5. The system displays an error message and returns to step 3.                                                                                                                                                     |

### Activity diagram:

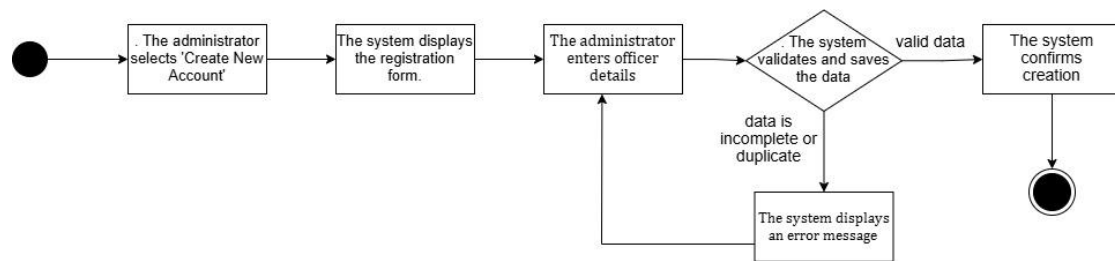


Figure 10 activity diagram “Create Police Account”

### sequence diagram:

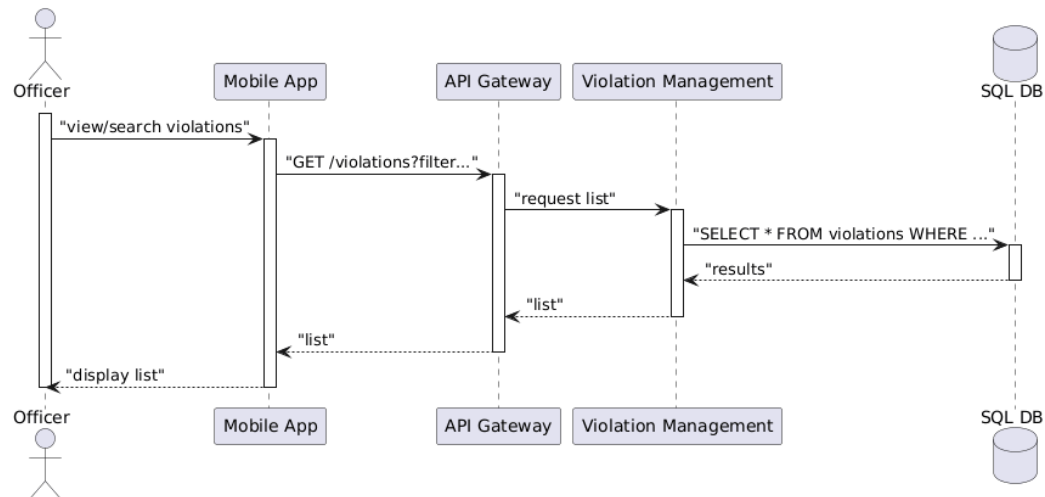


Figure 11 sequence diagram “Create Police Account”

### Uc -7-

#### use case specification:

|                         |                                                                                                                                                                                                                                    |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use case number</b>  | uc_7                                                                                                                                                                                                                               |
| <b>title</b>            | Edit Police Account                                                                                                                                                                                                                |
| <b>description</b>      | This feature allows the administrator to edit existing police officer account information.                                                                                                                                         |
| <b>actors</b>           | Administrator                                                                                                                                                                                                                      |
| <b>precondition</b>     | The administrator must be logged in.                                                                                                                                                                                               |
| <b>main flow</b>        | 1. The administrator selects 'Manage Police Accounts'.<br>2. The system displays the list.<br>3. The administrator selects an account.<br>4. The system displays account details.<br>5. The administrator edits and saves changes. |
| <b>postcondition</b>    | The police account information is updated successfully.                                                                                                                                                                            |
| <b>Alternative flow</b> | First alternative flow A1:<br>Start at step 6 where validation fails.<br>7. The system displays an error and requests correction.                                                                                                  |

### Uc -8-

#### use case specification:

|                        |                                                                                                                                                                                                                                          |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use case number</b> | Uc_8                                                                                                                                                                                                                                     |
| <b>Title</b>           | Activate / Deactivate Police Account                                                                                                                                                                                                     |
| <b>Description</b>     | This feature allows the administrator to enable or disable police officer accounts.                                                                                                                                                      |
| <b>Actors</b>          | Administrator                                                                                                                                                                                                                            |
| <b>Precondition</b>    | The administrator must be logged in.                                                                                                                                                                                                     |
| <b>Main Flow</b>       | 1. The administrator selects 'Manage Police Accounts'.<br>2. The system displays accounts.<br>3. The administrator selects an account.<br>4. The administrator confirms activation or deactivation.<br>5. The system updates the status. |
| <b>Postcondition</b>   | The selected police account is activated or deactivated successfully.                                                                                                                                                                    |

### Activity diagram:

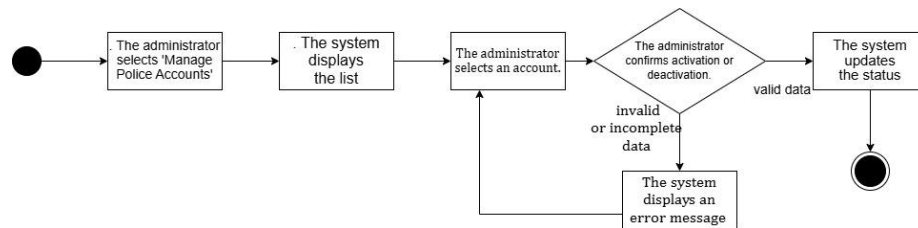


Figure 12 activity diagram “Activate / Deactivate Police Account”

### Uc -9-

#### use case specification:

|                  |                                                                                                                                                          |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use case number  | Uc_9                                                                                                                                                     |
| Title            | Review violation                                                                                                                                         |
| Description      | This feature allows the Police manager to review and audit violations submitted by police officers.                                                      |
| Actors           | Police manager                                                                                                                                           |
| Precondition     | The Police manager must be logged in.                                                                                                                    |
| Main Flow        | 1. The police manager selects 'Review Violations'.<br>2. The system displays the list.<br>3. The police manager searches and views a specific violation. |
| Postcondition    | The police manager has successfully reviewed the violation.                                                                                              |
| Alternative flow | First alternative flow A1:<br>Start at step 3 where no matching violation is found.<br>4. The system displays a message 'No matching violations found'.  |

## sequence diagram:

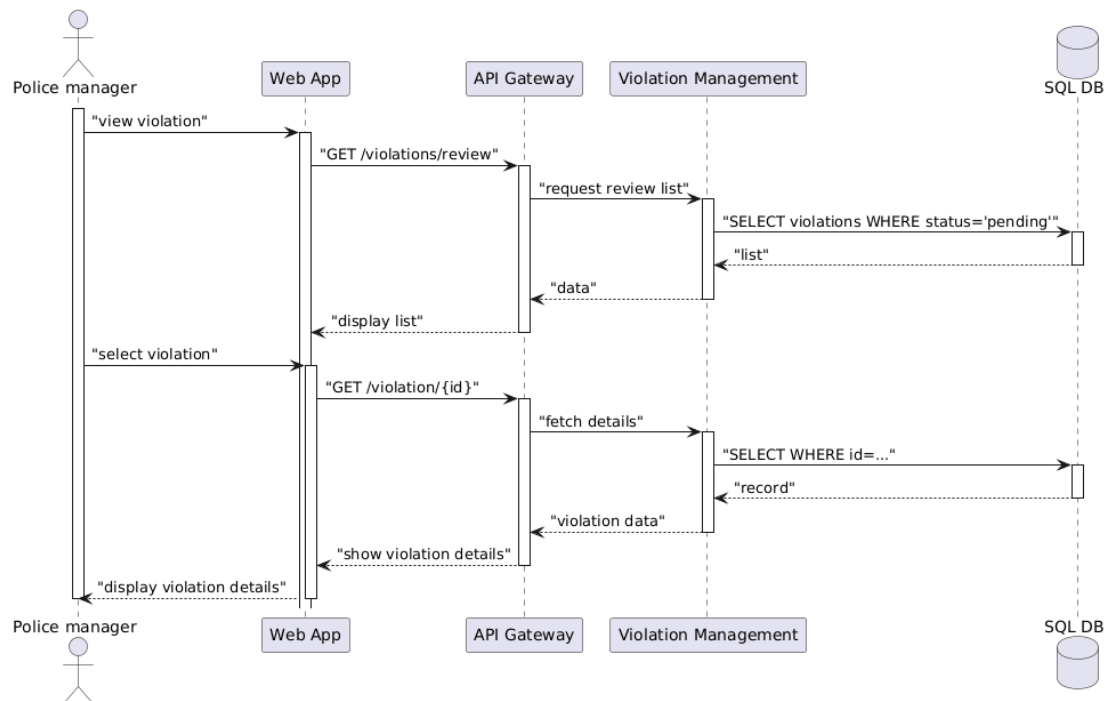


Figure 13 sequence diagram “Review violation”

## Uc -10-

### use case specification:

|                          |                                                                                                                                                                                                                                                                                                                                            |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use case number</b>   | Uc_10                                                                                                                                                                                                                                                                                                                                      |
| <b>Title</b>             | Review appeals                                                                                                                                                                                                                                                                                                                             |
| <b>Description</b>       | This feature allows the administrator to review and make decisions on citizen appeals.                                                                                                                                                                                                                                                     |
| <b>Actors</b>            | Police manager                                                                                                                                                                                                                                                                                                                             |
| <b>Precondition</b>      | The police manager must be logged in.                                                                                                                                                                                                                                                                                                      |
| <b>Main Flow</b>         | <ol style="list-style-type: none"><li>1. The police manager selects 'Review appeals.'</li><li>2. The system displays the list.</li><li>3. The police manager opens an appeal.</li><li>4. The system displays its details.</li><li>5. The police manager decides to accept or reject.</li><li>6. The system records the decision.</li></ol> |
| <b>Alternative flows</b> | First alternative flow A1:<br>Start at step 6 where decision submission fails.<br>7. The system displays 'Error submitting decision'.                                                                                                                                                                                                      |
| <b>Postcondition</b>     | The appeal status is updated successfully.                                                                                                                                                                                                                                                                                                 |

### sequence diagram:

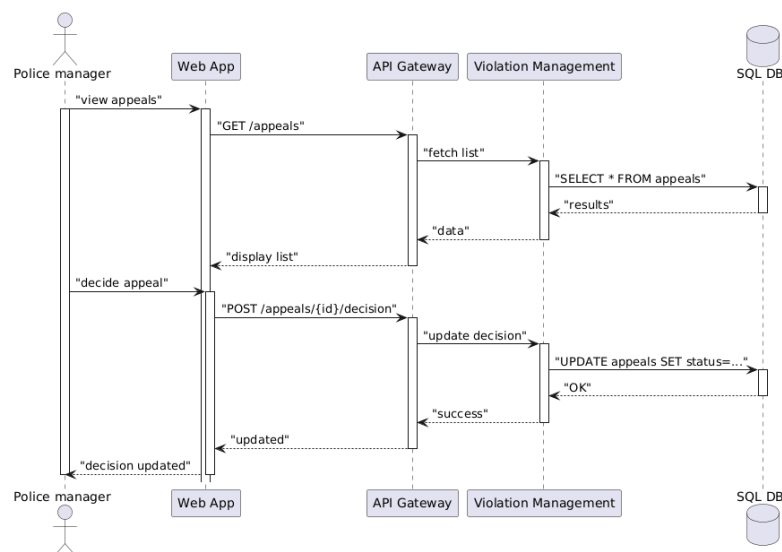


Figure 14 sequence diagram “Review appeals”



## Uc -11-

### use case specification:

|                          |                                                                                                                                                                                                                                                                            |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use case number</b>   | Uc_11                                                                                                                                                                                                                                                                      |
| <b>Title</b>             | View Violations                                                                                                                                                                                                                                                            |
| <b>Description</b>       | This feature allows citizens to view their traffic violations.                                                                                                                                                                                                             |
| <b>Actors</b>            | Citizen                                                                                                                                                                                                                                                                    |
| <b>Precondition</b>      | The citizen enters a valid vehicle number.                                                                                                                                                                                                                                 |
| <b>Main Flow</b>         | <ol style="list-style-type: none"><li>1. The citizen enters vehicle number.</li><li>2. The system verifies the data.</li><li>3. The system displays the violations list.</li><li>4. The citizen selects a violation.</li><li>5. The system displays its details.</li></ol> |
| <b>Alternative flows</b> | <p>First alternative flow A1:<br/>Start at step 2 where invalid data is entered.<br/>3. The system displays an error message.</p> <p>Second alternative flow A2:<br/>Start at step 3 where no violations exist.<br/>4. The system displays 'No violations found'.</p>      |
| <b>Postcondition</b>     | The citizen successfully views all related violations.                                                                                                                                                                                                                     |

## Uc -12-

### use case specification:

|                          |                                                                                                                                                                                                                                                                                      |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use case number</b>   | Uc_12                                                                                                                                                                                                                                                                                |
| <b>Title</b>             | Submit appeal                                                                                                                                                                                                                                                                        |
| <b>Description</b>       | This feature allows the citizen to submit an appeal on a specific violation.                                                                                                                                                                                                         |
| <b>Actors</b>            | Citizen                                                                                                                                                                                                                                                                              |
| <b>Precondition</b>      | The citizen must have viewed the violation beforehand.                                                                                                                                                                                                                               |
| <b>Main Flow</b>         | <ol style="list-style-type: none"><li>1. The citizen selects 'Submit appeal.</li><li>2. The system displays the appeal form.</li><li>3. The citizen enters the reason and notes.</li><li>4. The system validates and saves the appeal.</li></ol>                                     |
| <b>Alternative flows</b> | <p>First alternative flow A1:<br/>Start at step 4 where data is incomplete.<br/>5. The system displays 'Incomplete data'.</p> <p>Second alternative flow A2:<br/>Start at step 4 where a technical error occurs.<br/>6. The system displays 'Unable to submit appeal currently'.</p> |
| <b>Postcondition</b>     | The appeal is successfully submitted.                                                                                                                                                                                                                                                |

### sequence diagram:

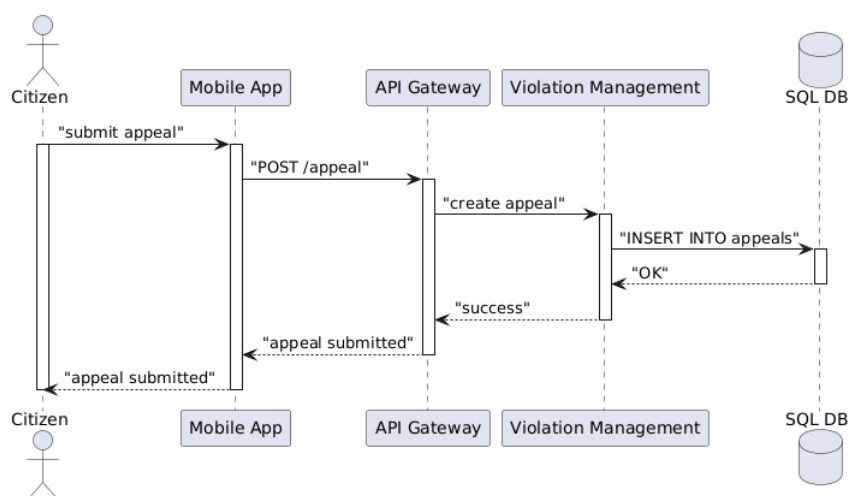


Figure 15 sequence diagram “Submit appeal”

## Class case diagram :

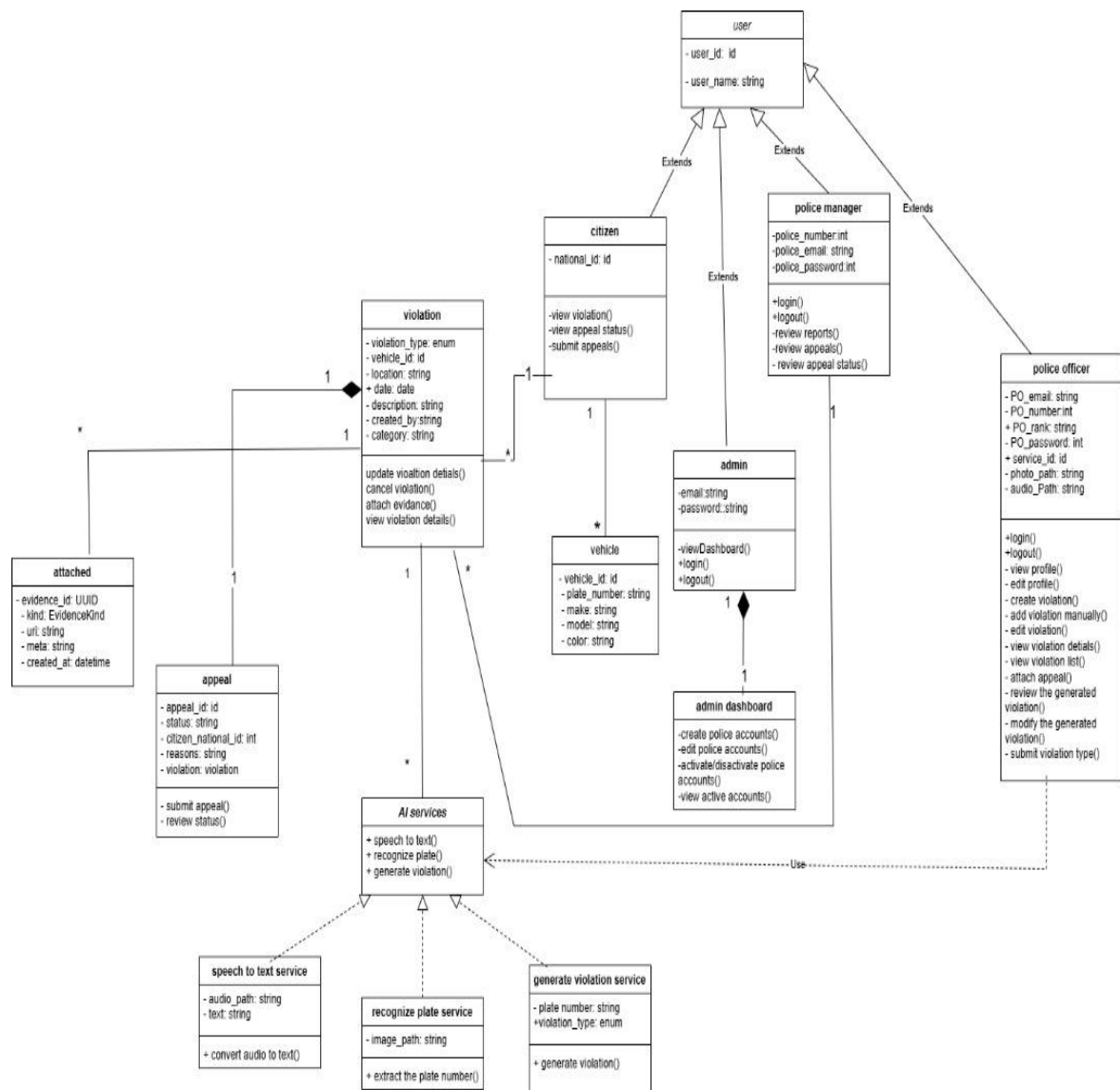


Figure 16 class diagram

## ERD – Entity Relationship Diagram :

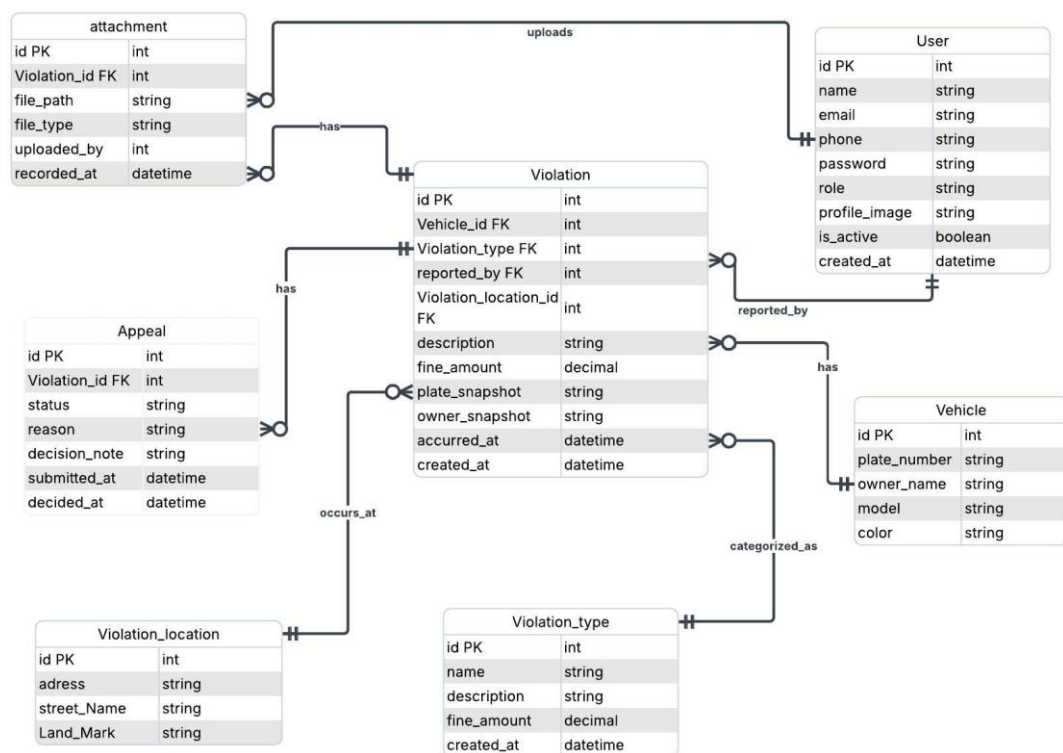


Figure 17 ERD

## Initial Test Cases:

**Test Case Scenario: Sce-01:** login/logout.

| Test case id | Test case title                                                    | Test steps                                                                                                       | Expected result                                                                                            |
|--------------|--------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| UC1-TC01     | Check successful login with valid credentials                      | 1. Launch the application on the login page.2. Enter a valid username and password.3. Select Login.              | Login is successful. The system grants access to the system main page. A user session is created.          |
| UC1-TC02     | Check login fails with missing fields                              | 1. Launch the application on the login page.2. Leave the username and/or password field empty.3. Select Login.   | The system displays an error message indicating a missing field and remains on the login page.             |
| UC1-TC03     | Check login fails with invalid credentials (non-existent username) | 1. Launch the application on the login page.2. Enter a non-existent username and any password.3. Select Login.   | The system displays an error message such as “Invalid username or password” and returns to the login form. |
| UC1-TC04     | Check login fails with invalid credentials (incorrect password)    | 1. Launch the application on the login page.2. Enter a valid username and an incorrect password.3. Select Login. | The system displays an error message such as “Invalid username or password” and returns to the login form. |
| UC1-TC05     | Check successful logout                                            | 1. Log in with valid credentials.2. Select Logout.                                                               | The user is logged out successfully. The session ends and the system returns to the login page.            |

**Test Case Scenario: Sce-02: Account Management:**

| Test case id | Test case title                                              | Test steps                                                                                                                          | Expected result                                                                                             |
|--------------|--------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| UC2-TC01     | Check view account profile                                   | 1. Log in as a police officer.2. Select View Profile.                                                                               | The system displays the officer's account details correctly.                                                |
| UC2-TC02     | Check successful edit of account information with valid data | 1. Log in as a police officer.2. Select View Profile.3. Select Edit Information.4. Enter valid new data.5. Select Save.             | The system validates and saves the data successfully and displays a success message.                        |
| UC2-TC03     | Check edit account information with invalid data             | 1. Log in as a police officer.2. Select View Profile.3. Select Edit Information.4. Enter invalid or incomplete data.5. Select Save. | The system displays an error message and returns to the edit information form.                              |
| UC2-TC04     | Check edit account information without making changes        | 1. Log in as a police officer.2. Select View Profile.3. Select Edit Information.4. Do not modify any fields.5. Select Save.         | The system keeps the existing data unchanged and displays an appropriate confirmation or no-change message. |

**Test Case Scenario: Sce-03: Generate violation:**

| Test case id | Test case title                                           | Test steps                                                                                                                                                                                                   | Expected result                                                                                                               |
|--------------|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| UC3-TC01     | Check successful generation of violation with valid data  | 1. Log in as a police officer.<br>2. Select Generate Violation.<br>3. Record audio successfully.<br>4. Capture an image of the vehicle.<br>5. Select violation type.<br>6. Review and confirm the violation. | The system converts the audio to text, extracts the license plate number, generates the violation, and saves it successfully. |
| UC3-TC02     | Check audio recording failure during violation generation | 1. Log in as a police officer.<br>2. Select Generate Violation.<br>3. Attempt to record audio but recording fails.                                                                                           | The system requests the officer to re-record the audio.                                                                       |
| UC3-TC03     | Check failure of audio-to-text conversion                 | 1. Log in as a police officer.<br>2. Select Generate Violation.<br>3. Record audio successfully.<br>4. Audio-to-text conversion fails.                                                                       | The system allows the officer to manually enter the violation text.                                                           |
| UC3-TC04     | Check license plate extraction failure                    | 1. Log in as a police officer.<br>2. Select Generate Violation.<br>3. Capture an unclear image of the vehicle.                                                                                               | The system requests the officer to manually enter the license plate number.                                                   |
| UC3-TC05     | Check reviewing generated violation                       | 1. Log in as a police officer.<br>2. Generate a violation.<br>3. Review the generated violation details.                                                                                                     | The system displays the generated violation details for review.                                                               |

#### Test Case Scenario: Sce-04: Add Violation.

| Test case id | Test case title                                          | Test steps                                                                                                                       | Expected result                                                                                      |
|--------------|----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| UC4-TC01     | Check successful addition of a violation with valid data | 1. Log in as a police officer.2. Select <b>Add Violation</b> .3. Enter valid violation details.4. Confirm the violation.         | The system validates the data, generates the violation report, and saves the violation successfully. |
| UC4-TC02     | Check add violation with missing required fields         | 1. Log in as a police officer.2. Select <b>Add Violation</b> .3. Leave one or more required fields empty.4. Submit the form.     | The system displays an error message and returns to the violation data entry form.                   |
| UC4-TC03     | Check add violation with invalid data                    | 1. Log in as a police officer.2. Select <b>Add Violation</b> .3. Enter invalid data (e.g., incorrect format).4. Submit the form. | The system displays an error message and allows the officer to correct the entered data.             |

#### Test Case Scenario: Sce-05: view violation.

| Test case id | Test case title                           | Test steps                                                                                                                                                             | Expected result                                                                                  |
|--------------|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| UC5-TC01     | Check viewing list of existing violations | 1. Log in as a police officer with existing violations. 2. Select <b>View Violations</b> .                                                                             | The system displays the list of existing violations successfully.                                |
| UC5-TC02     | Check viewing violation details           | 1. Log in as a police officer with existing violations.2. Select <b>View Violations</b> .3. Select view violation details.                                             | The system displays the detailed information of the selected violation.                          |
| UC5-TC03     | Check searching violations by date        | 1. Log in as a police officer with existing violations.2. Select <b>View Violations</b> .3. Search for violations using a date.4. Select a violation from the results. | The system displays the violations that match the selected date and shows the violation details. |
| UC5-TC04     | Check searching violations by             | 1. Log in as a police officer with existing violations.2. Select <b>View Violations</b> .3. Search for                                                                 | The system displays the violations that match the entered plate number and                       |



|                 |                                          |                                                                                              |                                                        |
|-----------------|------------------------------------------|----------------------------------------------------------------------------------------------|--------------------------------------------------------|
|                 | license plate number                     | violations using a license plate number.4. Select a violation from the results.              | shows the violation details.                           |
| <b>UC5-TC05</b> | Check viewing violations when none exist | 1. Log in as a police officer with no previous violations.2. Select <b>View Violations</b> . | The system displays the message “No violations found”. |

#### Test Case Scenario: Sce-06: Create Police Account

| Test case id    | Test case title                                               | Test steps                                                                                                                                                                               | Expected result                                                                                                            |
|-----------------|---------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>UC6-TC01</b> | Check successful creation of a police account with valid data | 1. Log in as an administrator.2. Select <b>Create New Account</b> .3. Enter valid and complete police officer details.4. Submit the registration form.                                   | The system validates the data, saves the new police account, and displays a successful creation confirmation.              |
| <b>UC6-TC02</b> | Check police account creation with missing required data      | 1. Log in as an administrator.2. Select <b>Create New Account</b> .3. Leave one or more required fields empty.4. Submit the registration form.                                           | The system displays an error message and returns to the registration form to correct the data.                             |
| <b>UC6-TC03</b> | Check police account creation with duplicate data             | 1. Log in as an administrator.2. Select <b>Create New Account</b> .3. Enter officer details that already exist (e.g., duplicate username or officer ID).4. Submit the registration form. | The system displays an error message indicating duplicate data and allows the administrator to re-enter valid information. |
| <b>UC6-TC04</b> | Check police account creation with invalid data               | 1. Log in as an administrator.<br>2. Select <b>Create New Account</b> .<br>3. Enter invalid data (e.g., incorrect email format, invalid ID format).<br>4. Submit the registration form   | The system displays an error message indicating invalid data and returns to the registration form for correction           |

### Test Case Scenario: Sce-07: edit police account

| Test case id | Test case title                                             | Test steps                                                                                                                                                                                                                                                                        | Expected result                                                                                                |
|--------------|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| UC7-TC01     | Check successful editing of police account with valid data. | <ol style="list-style-type: none"> <li>1. Log in as Administrator.</li> <li>2. Select <b>Manage Police Accounts</b>.</li> <li>3. Choose an existing police account from the list.</li> <li>4. Edit account details with valid data.</li> <li>5. Save the changes</li> </ol>       | The system validates the data and updates the police account successfully. A confirmation message is displayed |
| UC7-TC02     | Check editing police account with missing required fields   | <ol style="list-style-type: none"> <li>1. Log in as Administrator.</li> <li>2. Select <b>Manage Police Accounts</b>.</li> <li>3. Choose an existing police account.</li> <li>4. Remove or leave one or more required fields empty.</li> <li>5.</li> </ol>                         | The system displays an error message indicating missing required fields and requests correction                |
| UC7-TC03     | Check editing police account with invalid data format       | <ol style="list-style-type: none"> <li>1. Log in as Administrator.</li> <li>2. Select <b>Manage Police Accounts</b>.</li> <li>3. Choose an existing police account.</li> <li>4. Enter invalid data (e.g., incorrect phone number format).</li> <li>5. Save the changes</li> </ol> | The system rejects the input and displays a validation error message.                                          |

**Test Case Scenario: Sce-08: Activate / Deactivate Police Account :**

| Test case id | Test case title                                       | Test steps                                                                                                                                                                                                                                                      | Expected result                                                                                          |
|--------------|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| UC8-TC01     | Check successful activation of a police account       | <ol style="list-style-type: none"> <li>1. Log in as Administrator.</li> <li>2. Select <b>Manage Police Accounts</b>.</li> <li>3. Choose a deactivated police account.</li> <li>4. Confirm <b>Activate</b> action</li> </ol>                                     | The system activates the police account successfully and updates the account status to <i>Active</i>     |
| UC8-TC02     | Check successful deactivation of a police account     | <ol style="list-style-type: none"> <li>1. Log in as Administrator.</li> <li>2. Select <b>Manage Police Accounts</b>.</li> <li>3. Choose an active police account.</li> <li>4. Confirm <b>Deactivate</b> action</li> </ol>                                       | The system deactivates the police account successfully and updates the account status to <i>Inactive</i> |
| UC8-TC03     | Check activation/deactivation cancellation            | <ol style="list-style-type: none"> <li>1. Log in as Administrator.</li> <li>2. Select <b>Manage Police Accounts</b>.</li> <li>3. Choose any police account.</li> <li>4. Click <b>Activate/Deactivate</b>.</li> <li>5. Cancel the confirmation dialog</li> </ol> | The system cancels the operation and the police account status remains unchanged                         |
| UC8-TC04     | Check status update reflected in police accounts list | <ol style="list-style-type: none"> <li>1. Log in as Administrator.</li> <li>2. Select <b>Manage Police Accounts</b>.</li> <li>3. Activate or deactivate a police account.</li> <li>4. Return to the accounts list</li> </ol>                                    | The updated account status is correctly reflected in the police accounts list                            |

**Test Case Scenario: Sce-09: Review Violation:**

| Test case id | Test case title                                            | Test steps                                                                                                                                | Expected result                                                  |
|--------------|------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| UC9-TC01     | Check successful display of violations list                | 1. Log in as Police Manager.<br>2. Select Review Violations.                                                                              | The system displays the list of violations successfully.         |
| UC9-TC02     | Check successful search and review of a specific violation | 1. Log in as Police Manager.<br>2. Select Review Violations.<br>3. Search for an existing violation.<br>4. Select and view the violation. | The system displays the selected violation details successfully. |
| UC9-TC03     | Check system behavior when no matching violation is found  | 1. Log in as Police Manager.<br>2. Select Review Violations.<br>3. Search using invalid or non-existing criteria.                         | The system displays a message "No matching violations found".    |

**Test Case Scenario: Sce-10: Review Appeals:**

| Test case id | Test case title                                      | Test steps                                                                                                                                                                                     | Expected result                                                                         |
|--------------|------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| UC10-TC01    | Check display of appeals list                        | 1. Log in as Police Manager.<br>2. Select Review Appeals.                                                                                                                                      | The system displays the list of appeals successfully.                                   |
| UC10-TC02    | Check display of appeal details                      | 1. Log in as Police Manager.<br>2. Select Review Appeals.<br>3. Open an appeal from the list.                                                                                                  | The system displays the appeal details successfully.                                    |
| UC10-TC03    | Check successful acceptance of an appeal             | 1. Log in as Police Manager.<br>2. Select Review Appeals.<br>3. Open an appeal.<br>4. Review appeal details.<br>5. Select Accept.<br>6. Submit decision.                                       | The system records the decision successfully and updates the appeal status to Accepted. |
| UC10-TC04    | Check successful rejection of an appeal              | 1. Log in as Police Manager.<br>2. Select Review Appeals.<br>3. Open an appeal.<br>4. Review appeal details.<br>5. Select Reject.<br>6. Submit decision.                                       | The system records the decision successfully and updates the appeal status to Rejected. |
| UC10-TC05    | Check system behavior when decision submission fails | 1. Log in as Police Manager.<br>2. Select Review Appeals.<br>3. Open an appeal.<br>4. Review appeal details.<br>5. Select Accept or Reject.<br>6. Submit decision and a system failure occurs. | The system displays the message "Error submitting decision".                            |

### Test Case Scenario: Sce-11: view violation :

| Test case id | Test case title                                                  | Test steps                                                                                                                                                                                          | Expected result                                                                                    |
|--------------|------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| UC11-TC01    | Check successful viewing of violations with valid vehicle number | 1. Enter a valid vehicle number.<br>2. The system verifies the entered data.<br>3. The system displays the list of violations.<br>4. Select a violation from the list.<br>5. View violation details | The system displays the violations list and shows the selected violation details successfully      |
| UC11-TC02    | Check system behaviour with invalid vehicle number               | 1. Enter an invalid or incorrectly formatted vehicle number.<br>2. Submit the data.                                                                                                                 | The system displays an error message indicating invalid vehicle number and no violations are shown |
| UC11-TC03    | Check system behavior when no violations exist                   | 1. Enter a valid vehicle number with no associated violations.<br>2. Submit the data                                                                                                                | The system displays the message <b>"No violations found"</b>                                       |
| UC11-TC04    | Check display of violations list                                 | 1. Enter a valid vehicle number.<br>2. Submit the data                                                                                                                                              | The system displays a list of all violations related to the entered vehicle number                 |
| UC11-TC05    | Check display of violation details                               | 1. Enter a valid vehicle number.<br>2. View the violations list.<br>3. Select a violation.                                                                                                          | The system displays full details of the selected violation                                         |

### Test Case Scenario: Sce-12: submit Appeals:

| Test case id | Test case title                                                              | Test steps                                                                                                                                                                                                                                        | Expected result                                                                                        |
|--------------|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| UC12-TC01    | Check successful submission of an appeal with valid data                     | <ol style="list-style-type: none"> <li>1. View a specific violation.</li> <li>2. Select <b>Submit Appeal</b>.</li> <li>3. The system displays the appeal form.</li> <li>4. Enter valid reason and notes.</li> <li>5. Submit the appeal</li> </ol> | The system validates the data, saves the appeal successfully, and displays a confirmation message      |
| UC12-TC02    | Check submission of appeal with incomplete data                              | <ol style="list-style-type: none"> <li>1. View a specific violation.</li> <li>2. Select <b>Submit Appeal</b>.</li> <li>3. The appeal form is displayed.</li> <li>4. Leave required fields empty.</li> <li>5. Submit the appeal.</li> </ol>        | The system displays the message <b>"Incomplete data"</b> and the appeal is not saved                   |
| UC12-TC03    | Check system behavior when a technical error occurs during appeal submission | <ol style="list-style-type: none"> <li>1. View a specific violation.</li> <li>2. Select <b>Submit Appeal</b>.</li> <li>3. Enter valid appeal data.</li> <li>4. Submit the appeal.</li> <li>5. A technical/system error occurs</li> </ol>          | The system displays the message <b>"Unable to submit appeal currently"</b> and the appeal is not saved |
| UC12-TC04    | Check display of appeal submission form                                      | <ol style="list-style-type: none"> <li>1. View a specific violation.</li> <li>2. Select <b>Submit Appeal</b></li> </ol>                                                                                                                           | The system displays the appeal submission form correctly                                               |
| UC12-TC05    | Check that appeal is linked to the correct violation                         | <ol style="list-style-type: none"> <li>1. View a specific violation.</li> <li>2. Submit an appeal for that violation</li> </ol>                                                                                                                   | The system links the submitted appeal to the selected violation correctly                              |

### Initial RTM Requirements Trackability Matrix:

| Req-ID   | Use Case                                  | Analysis | System Design | Detailed Design | Coding | Test Case(s)         |
|----------|-------------------------------------------|----------|---------------|-----------------|--------|----------------------|
| RE-FR-01 | UC-1 (Login / Logout)                     | AM-AN    |               |                 |        | UC1-TC01, UC1-TC02   |
| RE-FR-02 | UC-1 (Login / Logout)                     | AM-AN    |               |                 |        | UC1-TC03             |
| RE-FR-03 | UC-2 (Account Management)                 | AM-AN    |               |                 |        | UC2-TC01, UC2-TC02   |
| RE-FR-04 | UC-3 (Generate Violation)                 | AM-AN    |               |                 |        | UC3-TC01             |
| RE-FR-05 | UC-4 (Add Violation)                      | AM-AN    |               |                 |        | UC4-TC01, UC4-TC02   |
| RE-FR-06 | UC-3 (Generate Violation – Voice)         | AM-AN    |               |                 |        | UC3-TC02, UC3-TC03   |
| RE-FR-07 | UC-3 (Review Generated Violation)         | AM-AN    |               |                 |        | UC3-TC04             |
| RE-FR-08 | UC-5 (Violation Management – Search/View) | BM-AN    |               |                 |        | UC5-TC01, UC5-TC02   |
| RE-FR-09 | UC-5 (Violation Management)               | BM-AN    |               |                 |        | UC5-TC03             |
| RE-FR-10 | UC-5 (Search Violation)                   | BM-AN    |               |                 |        | UC5-TC04             |
| RE-FR-11 | UC-11 (View Violations – Citizen)         | BM-AN    |               |                 |        | UC11-TC01            |
| RE-FR-12 | UC-12 (Submit Appeal)                     | BM-AN    |               |                 |        | UC12-TC01, UC12-TC02 |
| RE-FR-13 | UC-10 (Review Appeals)                    | BM-AN    |               |                 |        | UC10-TC01            |
| RE-FR-14 | UC-6 (Create Police Account)              | BM-AN    |               |                 |        | UC6-TC01             |



|          |                                          |       |  |  |  |           |
|----------|------------------------------------------|-------|--|--|--|-----------|
| RE-FR-15 | UC-6 (Create Police Account)             | BM-AN |  |  |  | UC6-TC02  |
| RE-FR-16 | UC-7 (Edit Police Account)               | BM-AN |  |  |  | UC7-TC01  |
| RE-FR-17 | UC-8 (Activate / Deactivate Account)     | BM-AN |  |  |  | UC8-TC01  |
| RE-FR-18 | UC-8 (Activate / Deactivate Account)     | BM-AN |  |  |  | UC8-TC02  |
| RE-FR-19 | UC-9 (Review Violations)                 | VM-AN |  |  |  | UC9-TC01  |
| RE-FR-20 | UC-9 (Search Reviewed Violations)        | VM-AN |  |  |  | UC9-TC02  |
| RE-FR-21 | UC-9 (View Violation Details)            | VM-AN |  |  |  | UC9-TC03  |
| RE-FR-22 | UC-10 (Review Appeals)                   | VM-AN |  |  |  | UC10-TC02 |
| RE-FR-23 | UC-10 (Accept Appeal)                    | VM-AN |  |  |  | UC10-TC03 |
| RE-FR-24 | UC-10 (Reject Appeal)                    | VM-AN |  |  |  | UC10-TC04 |
| RE-FR-25 | UC-3 (Voice to Text)                     | VM-AN |  |  |  | UC3-TC05  |
| RE-FR-26 | UC-3 (Image Analysis – Plate Extraction) | VM-AN |  |  |  | UC3-TC06  |
| RE-FR-27 | UC-3 (Automated Report Generation)       | VM-AN |  |  |  | UC3-TC07  |
| RE-FR-28 | UC-3 (Violation Classification)          | VM-AN |  |  |  | UC3-TC08  |

# **Chapter 5**

## **System Design**

## Introduction:

This chapter offers a comprehensive overview of the design phase steps, encompassing system design and architecture determination, as well as detailed design and modeling. It underscores the importance of these steps in creating a robust system for collecting, analyzing, and utilizing user feedback to continuously enhance the software system.

## System Architecture:

in this section we will discuss system design where system design involves defining the overall structure and architecture of the software, ensuring that all components work seamlessly together. This phase establishes the blueprint for efficient data flow, interaction, and integration within the system

**The first constraints to take into consideration** when building an architecture for a software system are the **non-functional requirements (quality attributes)**, so first we will start with the non-functional requirements of our system:

### 1. Usability:

The system shall provide intuitive and user-friendly interfaces that enable traffic police officers, police managers, and citizens to perform their tasks efficiently with minimal training. The workflows shall be simple and optimized for field operations, supported by clear visual cues such as meaningful icons and guided input forms to reduce user errors.

### 2. Security

The system shall protect sensitive data, including user account information, traffic violation records, and appeal data, by implementing secure authentication mechanisms, role-based access control, and encrypted communication channels. All data exchanges shall be secured using HTTPS protocols, and access to system resources shall be restricted according to user roles.

### 3. Performance

The system shall ensure acceptable response times for interactive operations such as login, data retrieval, and search functionalities. AI-based processes, including image analysis and speech-to-text conversion, shall be executed asynchronously to prevent degradation of system responsiveness.

### 4. Reliability

The system shall ensure reliable operation and data integrity under normal and failure conditions. Mechanisms such as error handling, data validation, and safe data persistence shall be implemented to prevent data loss during system or network failures.

### 5. Availability

The system shall be available during operational hours with minimal downtime. The design shall support continuous system operation to accommodate multiple concurrent users, including traffic police officers operating in the field.

### **Components functionalities:**

#### **API Gateway:**

Routes incoming API requests from client applications to the appropriate backend services and enforces access control.

#### **User Management:**

Manages user authentication, authorization, and account-related operations.

#### **Violation Management:**

Handles the creation, storage, retrieval, and management of traffic violation records.

#### **Speech-to-Text:**

Converts voice inputs from police officers into text for violation processing.

#### **Plate Recognition:**

Extracts vehicle plate numbers and related information from captured images.

#### **Violation Generation:**

Generates structured traffic violation reports by combining AI outputs and user input.

#### **Data Access:**

Provides standardized communication between system services and databases.

#### **SQL Database:**

Stores structured system data such as user accounts and violation records.

#### **NoSQL Database:**

Stores unstructured data generated by AI services, such as audio and image analysis results.

## System Architecture:

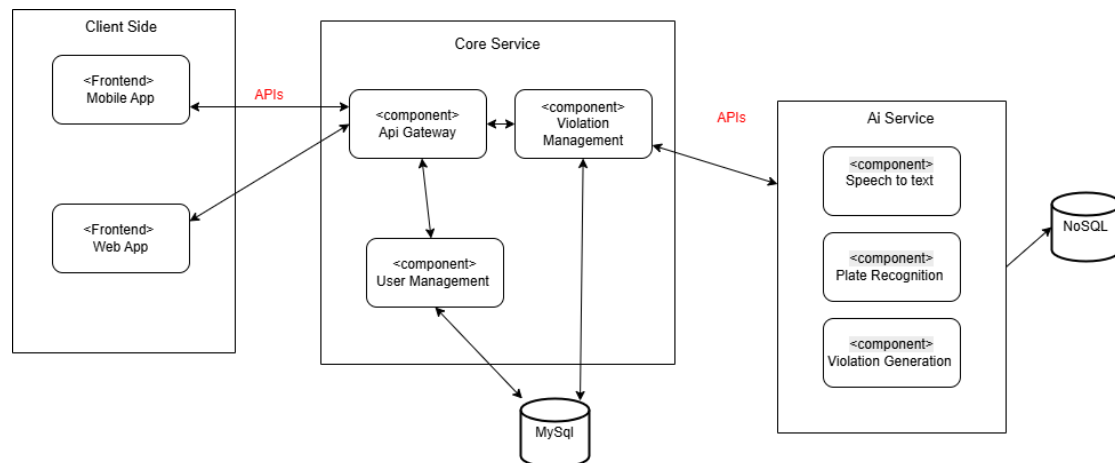


Figure 18 system architecture

The **Smart Traffic Police Assistant System** is designed using a **microservices architecture** combined with a **client-server model**, where system functionalities are divided into independent and loosely coupled services to improve scalability, maintainability, and fault isolation.

The **client side** consists of a mobile application and a web application that communicate with the backend through secured RESTful APIs.

The **backend** is composed of multiple independent microservices grouped into two main layers:

### Core Services:

This layer includes the **API Gateway**, **User Management**, and **Violation Management** services.

The API Gateway acts as a single entry point for all client requests and routes them to the appropriate microservices. The User Management service handles authentication and user account operations, while the Violation Management service manages traffic violation records and related operations.

### AI Services:

This layer contains specialized microservices responsible for advanced processing tasks, including **Speech-to-Text**, **Plate Recognition**, and **Violation Generation**. These services operate independently and communicate with the core services via APIs.

Each microservice manages its own data access. Structured system data such as user accounts and violation records are stored in a **MySQL database**, while AI-generated and unstructured data are stored in a **NoSQL database**.

## Database Design

SQL Database – MySQL:

In the **Smart Traffic Police Assistant System**, **MySQL** is used as the primary relational database to manage highly structured and transactional system data. This includes user accounts, traffic violation records, appeal data, roles, permissions, and system session information. MySQL provides strong data consistency, referential integrity, and efficient support for complex queries required for administrative, monitoring, and reporting functionalities.

The database schema is designed following normalization principles to reduce redundancy and enforce clear relationships between core entities, including:

**users :**

Stores user account information, including authentication credentials, role type (police officer, police manager, administrator, citizen), and account status.

**violations:**

Contains traffic violation records, including violation type, time, location, vehicle plate number, and status.

**appeals:**

Stores citizen appeal information linked to specific violations, including appeal reason, submission date, and decision status.

**roles and permissions:**

Manages role-based access control to ensure that system functionalities are accessed according to user privileges.

**sessions and system logs:**

Maintains session tracking, authentication states, and system activity logs for auditing and security purposes.

Foreign key constraints are applied to enforce data integrity across the system, such as linking **user\_id** in the violations table to the users table, and **violation\_id** in the appeals table to the violations table. This ensures consistent and reliable relationships between users, violations, and appeals.

## MySQL- database modeling:

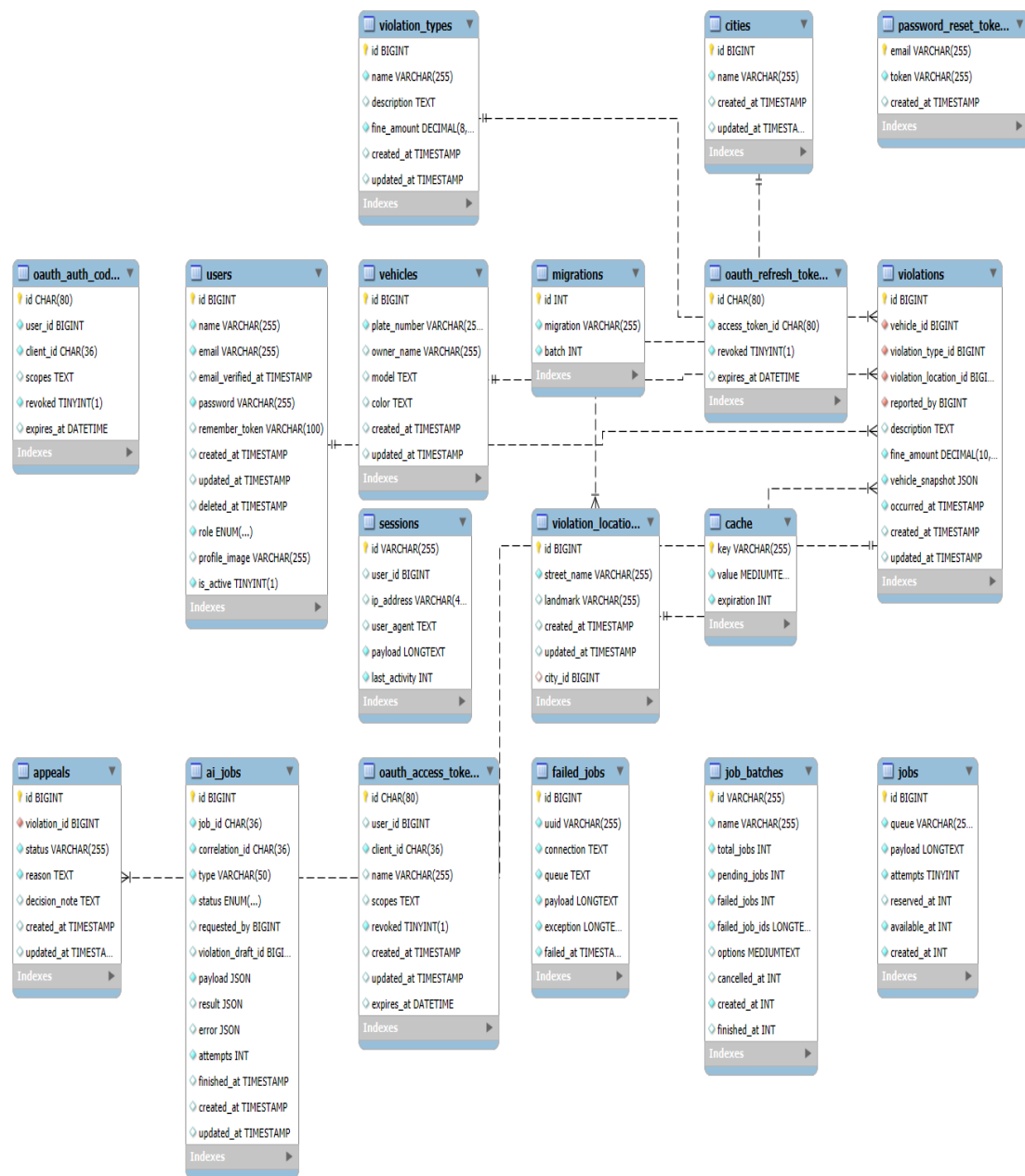


Figure 19 "MY\_SQL"

## Updating the RTM:

| Req-ID   | Use Case | Analysis | System Design             | Detailed Design | Coding | Test Case(s)       |
|----------|----------|----------|---------------------------|-----------------|--------|--------------------|
| RE-FR-01 | UC-1     | AM-AN    | User Management Service   | —               |        | UC1-TC01, UC1-TC02 |
| RE-FR-02 | UC-1     | AM-AN    | User Management Service   | —               |        | UC1-TC03           |
| RE-FR-03 | UC-2     | AM-AN    | User Management Service   | —               |        | UC2-TC01, UC2-TC02 |
| RE-FR-04 | UC-3     | AM-AN    | Violation + AI Services   | —               |        | UC3-TC01           |
| RE-FR-05 | UC-4     | AM-AN    | Violation Management      | —               |        | UC4-TC01, UC4-TC02 |
| RE-FR-06 | UC-3     | AM-AN    | Plate Recognition Service | —               |        | UC3-TC02, UC3-TC03 |
| RE-FR-07 | UC-3     | AM-AN    | Violation Management      | —               |        | UC3-TC04           |
| RE-FR-08 | UC-3     | AM-AN    | Violation Management      | —               |        | UC3-TC05           |
| RE-FR-09 | UC-5     | BM-AN    | Violation Management      | —               |        | UC5-TC01           |
| RE-FR-10 | UC-5     | BM-AN    | Violation Management      | —               |        | UC5-TC02           |
| RE-FR-11 | UC-11    | BM-AN    | Violation Management      | —               |        | UC11-TC01          |
| RE-FR-12 | UC-12    | BM-AN    | Appeal Management         | —               |        | UC12-TC01          |
| RE-FR-13 | UC-12    | BM-AN    | Appeal Management         | —               |        | UC12-TC02          |
| RE-FR-14 | UC-1     | AM-AN    | User Management Service   | —               |        | UC1-TC01           |



|          |       |        |                           |   |  |           |
|----------|-------|--------|---------------------------|---|--|-----------|
| RE-FR-15 | UC-1  | AM-AN  | User Management Service   | — |  | UC1-TC03  |
| RE-FR-16 | UC-2  | AM-AN  | User Management Service   | — |  | UC2-TC01  |
| RE-FR-17 | UC-6  | BM-AN  | User Management Service   | — |  | UC6-TC01  |
| RE-FR-18 | UC-7  | BM-AN  | User Management Service   | — |  | UC7-TC01  |
| RE-FR-19 | UC-7  | BM-AN  | User Management Service   | — |  | UC7-TC02  |
| RE-FR-20 | UC-8  | BM-AN  | User Management Service   | — |  | UC8-TC01  |
| RE-FR-21 | UC-6  | BM-AN  | User Management Service   | — |  | UC6-TC02  |
| RE-FR-22 | UC-10 | DM-AN  | Appeal Management         | — |  | UC10-TC01 |
| RE-FR-23 | UC-10 | DM-AN  | Appeal Management         | — |  | UC10-TC02 |
| RE-FR-24 | UC-10 | DM-AN  | Appeal Management         | — |  | UC10-TC03 |
| RE-FR-25 | UC-3  | FGM-AN | AI – Speech to Text       | — |  | UC3-TC06  |
| RE-FR-26 | UC-3  | FGM-AN | AI – Plate Recognition    | — |  | UC3-TC07  |
| RE-FR-27 | UC-3  | FGM-AN | AI – Violation Generation | — |  | UC3-TC08  |
| RE-FR-28 | UC-3  | FGM-AN | AI – Classification       | — |  | UC3-TC09  |

# **Chapter 6**

## **AI service**

## 1. Introduction

Artificial Intelligence (AI) is playing an increasingly important role in modern information systems by automating complex tasks, improving data accuracy, and enhancing decision-making processes. In the domain of traffic management and law enforcement, traffic police officers are required to document violations accurately while operating in dynamic and often challenging field environments. Manual data entry during traffic enforcement is time-consuming, error-prone, and can distract officers from their primary duties.

The AI module in the Traffic Police Assistant System is designed to address these challenges by automating the extraction of critical information from unstructured inputs such as spoken audio and captured images. By leveraging advanced Speech-to-Text (STT) and computer vision technologies, the system enables traffic police officers to record violations efficiently and accurately without relying on manual typing.

Designed as an independent and modular component, the AI service enhances the overall efficiency, accuracy, and reliability of traffic violation documentation, supporting smarter and faster law enforcement workflows.

## 2. Key Functionalities of the AI Module

The AI module processes traffic violation data through multiple intelligent components, each responsible for handling a specific type of input:

- **Speech-to-Text (STT):**  
Converts the spoken description of traffic violations recorded by the police officer into structured textual data. The STT component supports the Arabic language and is designed to handle dialectal variations and real-world noise conditions.
- **License Plate Recognition (Vision & OCR):**  
Extracts vehicle license plate numbers from captured images using computer vision and text recognition models. This component ensures accurate identification of vehicles involved in traffic violations.
- **Data Structuring and Validation:**  
Organizes the extracted textual information into structured fields such as violation type, vehicle details, location, and additional notes, enabling efficient storage and further processing within the system.

In this chapter, we present the core AI technologies used in the Traffic Police Assistant System, review related work and existing models, and describe the design, implementation, and experimental evaluation of the proposed AI components.

### 3. Basic Concepts :

#### 1. General Concept of Artificial Intelligence in the System

The Traffic Police Assistant System relies on Artificial Intelligence (AI) techniques to automate the processing of unstructured data, such as audio and images, generated during field operations. The main objective of the AI component is to transform this unstructured input into structured and usable textual information that can support traffic violation documentation and decision-making.

Conceptually, AI in this system aims to replicate selected human capabilities, such as hearing and visual perception. The system is designed to understand what the traffic police officer says and analyze what the camera captures, without requiring extensive manual data entry.

#### 2. Core AI Components of the System

From a high-level perspective, the AI module of the proposed system is divided into two main components that operate independently but complement each other:

##### Speech Processing Component

This component focuses on converting the spoken description of a traffic violation recorded by the police officer into written text using Speech-to-Text (STT) technologies. This approach enables faster documentation, reduces manual effort, and allows hands-free interaction in real-world traffic environments.

##### Vision-Based Processing Component

This component is responsible for extracting visual information from captured images, particularly vehicle license plate numbers. It relies on computer vision techniques and Optical Character Recognition (OCR) or vision-language models to analyze images and extract textual information accurately.

#### 3. Speech-to-Text Concept

Speech-to-Text is an AI application that converts human speech signals into corresponding text. This process involves analyzing audio signals, extracting acoustic features, and mapping them to linguistic representations using machine learning and deep learning models.

In the context of the Traffic Police Assistant System, STT is used to convert spoken traffic violation descriptions into official textual records. However, Arabic speech recognition introduces additional challenges, such as dialectal variations,

pronunciation differences, and environmental noise, which must be carefully considered when selecting an appropriate STT model.

#### **4. Computer Vision and Text Extraction Concept**

Computer vision refers to the ability of intelligent systems to interpret and analyze visual content. In this project, vision-based techniques are employed to extract license plate numbers from vehicle images captured at the scene of the violation.

This process can be implemented using a traditional two-stage pipeline, consisting of license plate detection followed by OCR, or through modern vision-language models that integrate detection and text extraction within a single framework.

#### **5. Integration of AI Components**

The integration of speech-based and vision-based AI components allows the Traffic Police Assistant System to generate a comprehensive and reliable traffic violation record. By combining descriptive information obtained from speech with objective visual data extracted from images, the system improves data accuracy, reduces human error, and enhances overall operational efficiency.

## 4. Literature review:

### AI Literature Review

#### License Plate Recognition and OCR Models

##### 1. Introduction

Vehicle License Plate Recognition (LPR) is a critical component in intelligent traffic enforcement systems. The task aims to automatically identify vehicle license numbers from images captured in real-world environments. From a conceptual perspective, the LPR process consists of two main stages: detecting the license plate region within an image and extracting the alphanumeric characters from that region.

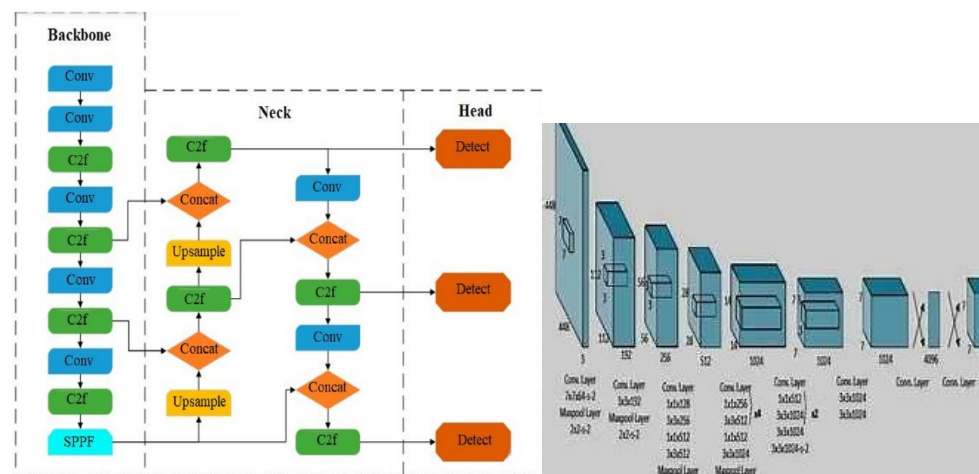
Traditionally, this task has been addressed using a two-stage pipeline that combines object detection models with Optical Character Recognition (OCR) techniques. While effective in controlled scenarios, this approach increases system complexity and introduces error propagation between detection and recognition stages.

Recent advances in vision-language models have introduced an alternative unified approach, where detection and text extraction are handled within a single model. This study evaluates both traditional and modern approaches to identify the most suitable solution for the Traffic Police Assistant System.

##### 2. Traditional License Plate Recognition Pipeline

###### 2.1 YOLOv8 for License Plate Detection

YOLOv8 is a real-time object detection model widely used for detecting objects such as vehicles and license plates. In license plate recognition systems, YOLOv8 is responsible for localizing the license plate region, which is then passed to an OCR model for character extraction.



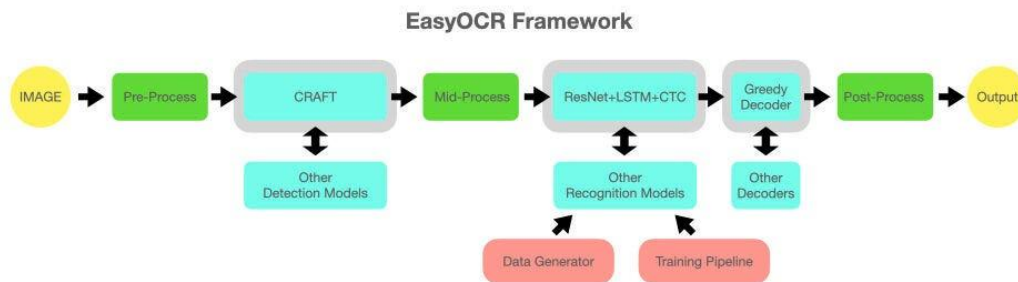
Although YOLOv8 provides fast and accurate detection, the overall system performance depends heavily on the accuracy of the subsequent OCR stage.

## 2.2 OCR Models for Text Extraction

Once the license plate is detected, OCR models are used to extract the alphanumeric characters. In this study, EasyOCR and Tesseract OCR are evaluated as traditional OCR solutions.

### 2.2.1 EasyOCR

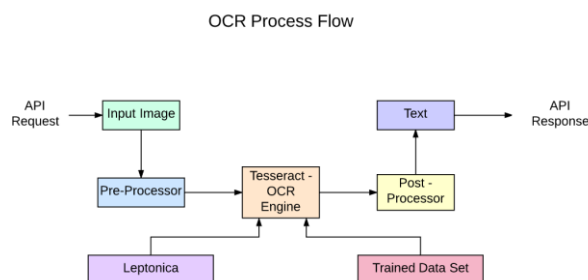
EasyOCR is an open-source OCR framework based on deep learning architectures. It supports multiple languages and is widely used for general OCR tasks.



However, EasyOCR struggles with license plate images due to small character size, varying lighting conditions, and mixed Arabic–English digits, often resulting in partial or incorrect recognition.

### 2.2.2 Tesseract OCR

Tesseract OCR is one of the most widely adopted open-source OCR engines, originally developed for document-style text recognition.

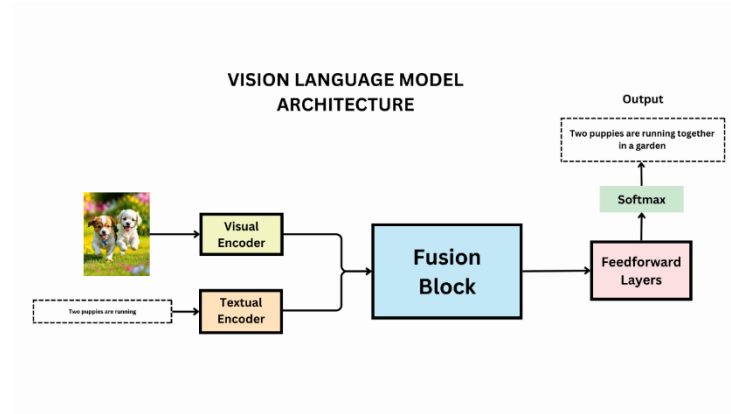


Despite its success in document processing, Tesseract performs poorly in scene text recognition tasks such as license plates, where noise, perspective distortion, and complex backgrounds are present.

### 3. Vision-Language Model Approach

#### 3.1 Qwen3 VL:2b

Qwen3 VL:2b is a modern vision-language model designed to jointly process visual and textual information. Unlike traditional pipelines, it can directly generate textual output from an image without requiring a separate detection or OCR stage.



This unified architecture significantly reduces pipeline complexity and minimizes error propagation, making Qwen3 VL:2b particularly suitable for license plate recognition in complex real-world scenarios.

### 4. Experimental Evaluation

#### 4.1 EasyOCR (Without Detection)



#### Ground Truth OCR Output Accuracy CER

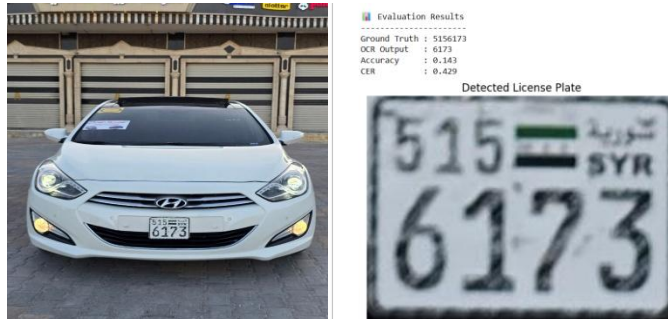
|         |         |       |       |
|---------|---------|-------|-------|
| 5108703 | 5855103 | 0.429 | 0.571 |
|---------|---------|-------|-------|

The results indicate that EasyOCR alone fails to extract complete license plate numbers accurately.



## 4.2 EasyOCR with YOLOv8

Combining EasyOCR with YOLOv8 improves plate localization but does not resolve incomplete character recognition. In many cases, only part of the license plate number is extracted.



## 4.3 Qwen3 VL:2b Results



### Example Ground Truth Output Accuracy CER

|   |         |         |      |     |
|---|---------|---------|------|-----|
| 1 | 5077237 | 5077237 | 1.00 | 0.0 |
| 2 | 5103682 | 5103682 | 1.00 | 0.0 |
| 3 | 233622  | 233622  | 1.00 | 0.0 |

Qwen3 VL:2b achieved perfect recognition across all tested cases.

## 4.4 Tesseract OCR Results

Tesseract OCR failed to produce reliable results both independently and when combined with YOLOv8, confirming its limitations in real-world license plate recognition.

## 5. Summary of Key Findings

EasyOCR produced high character error rates and incomplete outputs.

EasyOCR combined with YOLOv8 improved localization but not recognition accuracy.

Tesseract OCR was unsuitable for scene text recognition tasks.

Qwen3 VL:2b achieved superior accuracy and zero CER while simplifying the pipeline.

## 6. Conclusion

Based on experimental results, Qwen3 VL:2b was selected as the primary license plate recognition model for the Traffic Police Assistant System due to its high accuracy, robustness, and integrated architecture.

## Speech-to-Text (STT) Models

### 1. Introduction

Speech-to-Text (STT) technology plays a critical role in intelligent systems that rely on spoken input to generate structured textual data. In traffic enforcement environments, police officers often operate under time pressure and challenging conditions, making manual text entry inefficient and error-prone. STT provides an effective solution by enabling officers to verbally describe traffic violations while the system automatically converts speech into text.

Within the Traffic Police Assistant System, the STT component is responsible for transforming spoken violation descriptions—including vehicle details, locations, and violation types—into accurate textual records. Due to the linguistic complexity of Arabic, environmental noise, and dialectal variations, selecting a robust and reliable STT model is essential.

This study presents a comparative evaluation of state-of-the-art STT models, focusing on their suitability for real-world deployment in traffic enforcement scenarios.

### 2. Overview of Speech-to-Text Technology

Speech-to-Text systems aim to convert human speech signals into written text by analyzing acoustic patterns and mapping them to linguistic representations. Modern STT systems are primarily based on deep learning architectures, particularly Transformer models, which have demonstrated strong performance in multilingual and noisy environments.

An STT pipeline typically involves:

Audio preprocessing and feature extraction

Acoustic modeling

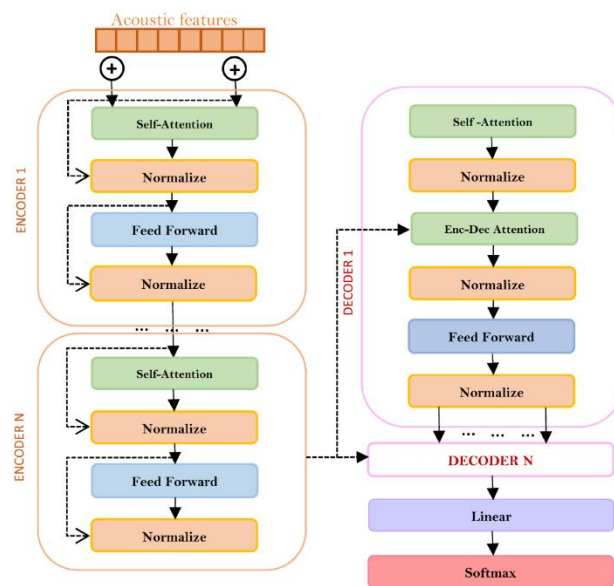
Language modeling

Text decoding

### 3.1 Whisper

Whisper is an automatic speech recognition system developed by OpenAI and trained on approximately 680,000 hours of multilingual and multitask supervised data collected from the web. This large-scale and diverse dataset enables Whisper to achieve high robustness against accents, background noise, and technical vocabulary.

Whisper follows an end-to-end encoder–decoder Transformer architecture. Input audio is segmented into 30-second chunks, converted into log-Mel spectrograms, and passed through the encoder. The decoder then generates the corresponding text output, guided by special tokens that support tasks such as language identification, transcription, and translation.



Whisper is available in multiple sizes (tiny, base, small, medium, large), allowing trade-offs between accuracy, processing speed, and memory usage.

### 3.2 Vosk (Arabic-Supported STT)

Vosk is an open-source offline speech recognition toolkit designed for lightweight and real-time applications. It supports multiple languages, including Arabic, and can operate without an internet connection, making it suitable for privacy-sensitive environments.

Vosk relies on traditional acoustic and language modeling approaches optimized for low resource consumption. While it offers fast inference and low memory usage, its recognition accuracy—particularly for Arabic dialects—is often lower than modern Transformer-based models.

## 4. Comparison Criteria

To objectively evaluate the STT models, the following comparison criteria were defined prior to experimentation:

Word Error Rate (WER): Measures transcription errors at the word level

Accuracy Percentage: Calculated as  $(1 - \text{WER}) \times 100$

Processing Time: Time required to transcribe a single audio file

Memory Consumption (RAM): Runtime memory usage

Language Support: Ability to accurately transcribe Arabic speech

Offline Capability: Suitability for deployment without internet access

Robustness: Performance under noisy and real-world conditions

## **5. Experimental Setup**

All experiments were conducted in a local offline environment to simulate real-world deployment conditions for traffic police officers.

Experimental setup included:

Audio recordings of traffic violation descriptions

Arabic language recordings by native speakers

Realistic background noise conditions

Ground truth transcripts manually prepared for evaluation

WER computation using standard evaluation tools

Two Whisper model variants (Small and Medium) were evaluated and compared with Vosk for Arabic speech recognition.

## **6. Experimental Results**

### **6.1 Whisper – Small Model**

Processing Time: ~23 seconds per audio

Accuracy: ~30%

Observations:

Fast execution

Low memory usage

High transcription errors in Arabic

### **6.2 Whisper – Medium Model**

Processing Time: ~105 seconds per audio

Accuracy: ~70%

Observations:

Significantly improved accuracy

Better handling of Arabic pronunciation

Higher memory consumption

### **6.3 Vosk (Arabic)**

Processing Time: Very fast

Accuracy: Low

Observations:

Supports Arabic offline

Fails to capture full semantic meaning

Not suitable for detailed violation descriptions

### **7. Discussion**

The experimental results demonstrate a clear trade-off between model size, accuracy, and computational cost. While Whisper Small offers faster inference, its transcription accuracy for Arabic speech is insufficient for reliable traffic violation documentation. Whisper Medium, despite higher processing time and memory requirements, provides a substantially better transcription quality.

Vosk, although lightweight and offline-capable, fails to meet the accuracy requirements necessary for legal and administrative traffic enforcement tasks.

The superior performance of Whisper Medium can be attributed to its larger model capacity and extensive multilingual training data, which enable better generalization across Arabic speech patterns.

### **8. Conclusion**

Based on the comparative evaluation, Whisper Medium was selected as the primary Speech-to-Text model for the Traffic Police Assistant System. It provides the best balance between transcription accuracy and robustness for Arabic speech, despite higher computational requirements.

Future work may explore optimization techniques, model quantization, or GPU acceleration to improve inference speed while maintaining accuracy.

## Service Design and Implementation

### 3.1 Introduction

In this section, we present the design and implementation of the Artificial Intelligence (AI) service used in the Traffic Police Assistant System. The AI service is designed to efficiently process unstructured inputs generated during traffic enforcement operations, particularly audio recordings and captured images, and transform them into structured and reliable traffic violation records.

The service follows a modular and scalable architecture that enables independent processing of Speech-to-Text (STT) and License Plate Recognition components. By leveraging advanced deep learning models and well-defined processing stages, the AI service supports accurate transcription of spoken violation descriptions, extraction of vehicle information from images, and organization of the extracted data into a structured format suitable for storage and further analysis.

### 3.2 AI Service Design

The AI service is designed as an independent processing component that can be integrated with external systems through well-defined APIs. Its architecture supports asynchronous processing, allowing traffic officers to submit violation data without waiting for immediate results, while the system processes inputs in the background.

The service pipeline consists of multiple stages, each responsible for a specific task, including audio transcription, image-based license plate recognition, and data structuring. This design ensures flexibility, maintainability, and ease of future expansion.

#### 3.2.1 Service Functional Requirements

The functional requirements of the AI service are defined as follows:

The service must support receiving audio recordings of traffic violation descriptions recorded by traffic police officers.

The service must support receiving vehicle images captured during traffic enforcement for license plate recognition.

The service must be capable of retrieving audio and image files from external storage sources (e.g., cloud storage or local repositories) for processing.

The service must convert received audio files into accurate textual transcripts using the Speech-to-Text (STT) component.

The service must support Arabic language transcription and handle realistic field conditions, including background noise and pronunciation variations.

The service must extract vehicle license plate numbers from captured images using computer vision and OCR or vision-language models.

The service must organize the extracted information into structured fields, including violation description, vehicle details, location, and additional notes.

The service must provide an API interface that allows external systems or applications to submit data for processing.

The service must track and store the processing status of each request at different stages (e.g., STT processing, image analysis, data structuring).

The service must provide an API endpoint to retrieve the real-time processing status of submitted requests.

The service must provide an API endpoint to retrieve the final structured traffic violation record, including transcribed text and extracted license plate information.

### **Non-Functional Requirements :**

#### **1) Performance**

- The AI Service must process jobs asynchronously; STT/OCR processing time depends on file size but must be measured and monitored.
- The system must enforce file size limits (e.g., 25–50MB) and return clear validation messages.

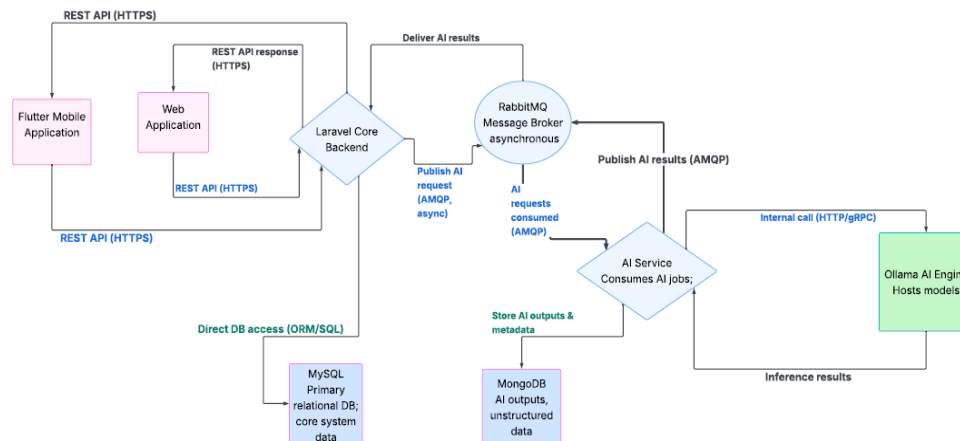
#### **2) Scalability**

The system must support horizontal scaling by increasing RabbitMQ consumers.  
The AI Service should allow multiple instances/workers (STT/OCR) without conflicts.

#### **3) Reliability**

- No job loss: every job published to RabbitMQ must be processed or retried safely.  
Implement a retry policy with a maximum attempt count and a Dead Letter Queue (DLQ) for failures.
- Persist job states (Queued / Processing / Done / Failed) in metadata storage.

## Processing Flow:



**Figure : AI Service Process Flow in the Traffic Police Assistant System**

This figure illustrates the end-to-end processing flow. Client applications (Flutter/Web) submit requests to the Laravel backend via REST APIs. The backend publishes AI jobs (STT/OCR) asynchronously to RabbitMQ. The AI Service consumes these jobs, calls the model engine (Ollama) for inference, and stores AI outputs and metadata in MongoDB, while core system data remains in MySQL. Finally, Laravel retrieves the results and exposes them back to clients through API endpoints.

## AI Processing Steps

The AI service processes traffic violation data through a structured multi-stage pipeline. The processing steps are as follows:

- **Audio Acquisition:**  
Receive recorded audio files from the mobile application or retrieve them from external storage through secure API endpoints.
- **Audio Preprocessing:**  
Normalize the audio format, sampling rate, and channels to ensure compatibility with the Speech-to-Text model.
- **Speech-to-Text Conversion:**  
Convert the processed audio input into textual form using the selected Speech-to-Text model, producing a raw transcription of the traffic violation description.
- **Text Post-processing:**  
Clean and normalize the generated text by correcting spacing, removing noise tokens, and standardizing numbers and entities (e.g., vehicle numbers, locations).



- **Image Acquisition (OCR Path):**  
Receive vehicle images captured by the traffic officer or retrieved from storage for license plate recognition.
- **License Plate Recognition:**  
Extract the license plate number either through a traditional detection and OCR pipeline (YOLOv8 + OCR) or through a unified vision-language model.
- **Result Validation:**  
Validate the extracted textual outputs by checking confidence scores, format consistency, and language correctness.
- **Data Storage:**  
Store structured AI outputs and metadata in the database for later retrieval and auditing.
- **Result Delivery:**  
Return the final transcription and recognition results to the backend service through asynchronous message communication.

## Used Tools and Models

This section presents the main tools, frameworks, and artificial intelligence models used in the development and implementation of the AI service within the Traffic Police Assistant System. These tools were selected based on their reliability, performance, and suitability for real-world traffic enforcement scenarios.

### 1. Backend and Integration Tools

#### **Laravel Framework**

Laravel is used as the core backend framework responsible for handling API requests, managing authentication, orchestrating AI processing jobs, and communicating with other system components. It provides a clean architecture, robust security features, and seamless integration with databases and message queues.

#### **RESTful APIs (HTTPS)**

REST APIs are used to enable communication between client applications (Flutter and Web) and the backend services. All APIs use HTTPS to ensure secure data transmission.

#### **RabbitMQ (Message Broker)**

RabbitMQ is used as an asynchronous message broker to manage AI processing jobs. It decouples the backend from the AI service, allowing scalable and fault-tolerant job processing. Each AI task (STT or OCR) is published as a message and consumed by the AI service workers.

## **2. Databases**

### **MongoDB**

MongoDB is used to store unstructured AI outputs and metadata, including transcription results, OCR outputs, confidence scores, and processing logs. Its flexible schema makes it suitable for handling AI-generated data.

## **3. Speech-to-Text (STT) Models and Tools**

### **Whisper**

Whisper is used as the primary Speech-to-Text model. The Medium version was selected due to its superior accuracy in Arabic speech recognition compared to smaller variants. Whisper converts spoken traffic violation descriptions into structured text.

### **Python**

Python is used to implement the STT processing logic and model execution due to its strong ecosystem for machine learning and audio processing.

## **4. License Plate Recognition and OCR Models**

### **Qwen3 VL:2b**

Qwen3 VL:2b is the primary vision-language model used for license plate recognition. It performs image-to-text extraction directly, eliminating the need for a separate detection and OCR pipeline.

## **5. AI Inference Engine**

### **Ollama AI Engine**

Ollama is used to host and run AI models locally, including vision-language models. It enables on-premise inference, enhances data privacy, and reduces dependency on external cloud services.

## **6. Frontend Tools**

### **Flutter Mobile Application**

Flutter is used to develop the mobile application for traffic police officers, allowing them to record audio, capture images, and submit violations.

### **Web Application**

A web-based interface is provided for administrative review, monitoring AI processing status, and accessing structured violation reports.

# **Chapter 7**

## **Practical Implementation**

## Introduction:

In this chapter, we will present the practical implementation of the system, detailing the technologies and tools employed. Additionally, we will showcase the system interfaces and conclude by executing test cases to verify the system's functionalities across different scenarios.

## Used Tools:

- **Laravel Framework**

Laravel is a powerful PHP web framework designed to simplify the development of robust and scalable web applications. It follows the MVC (Model–View–Controller) architecture and provides built-in features such as routing, authentication, validation, and database abstraction. In our system, Laravel is used as the core backend framework responsible for handling RESTful APIs, managing user requests, orchestrating AI processing workflows, and integrating with databases and message queues.

- **Flutter**

Flutter is an open-source UI framework developed by Google for building cross-platform applications from a single codebase. It enables the development of high-performance mobile applications with a rich and responsive user interface. In this project, Flutter is used to build the mobile application used by traffic police officers to record audio, capture vehicle images, and submit traffic violations efficiently.

- **RabbitMQ**

RabbitMQ is an open-source message broker that implements asynchronous message queuing. It enables reliable communication between distributed system components. In our system, RabbitMQ is used to manage AI processing jobs asynchronously, allowing the backend to send Speech-to-Text and OCR requests to the AI service without blocking, thus improving scalability and system performance.

- **Python**

Python is a high-level programming language widely used in artificial intelligence and machine learning applications. It offers a rich ecosystem of libraries and frameworks for audio processing, computer vision, and natural language processing. In this project, Python is used to implement the AI service, including Speech-to-Text processing and OCR-related tasks.

- **Whisper (Speech-to-Text Model)**

Whisper is an automatic speech recognition (ASR) model developed by OpenAI. It is trained on large-scale multilingual datasets and demonstrates strong robustness to noise, accents, and different speaking styles. The Whisper Medium model is used in this system to convert spoken traffic violation descriptions into accurate textual transcripts, with strong support for the Arabic language.

- **Qwen3 VL:2b**

Qwen3 VL:2b is a modern vision-language model capable of processing images and

generating textual outputs directly. Unlike traditional OCR pipelines, it can perform image-to-text recognition without requiring a separate object detection stage. In this project, Qwen3 VL:2b is used as the primary model for license plate recognition due to its high accuracy and simplified processing workflow.

- **Ollama AI Engine**

Ollama is a local AI inference engine used to host and run large language and vision-language models on-premise. It enables efficient and privacy-preserving execution of AI models without relying on external cloud services. In this system, Ollama is used to run vision-language models for OCR tasks.

- **MySQL Database**

MySQL is an open-source relational database management system widely used for structured data storage. It is known for its reliability, scalability, and performance. In this project, MySQL is used to store core system data such as users, vehicles, traffic violations, and AI processing references.

- **MongoDB**

MongoDB is a NoSQL database designed for storing unstructured and semi-structured data. It provides flexible schemas and high scalability. In this system, MongoDB is used to store AI-generated outputs, including STT transcripts, OCR results, confidence scores, and processing metadata.

- **Postman**

Postman is an API testing and development tool that simplifies the process of designing, testing, and debugging RESTful APIs. It is used extensively during development to validate backend endpoints and AI service integrations.

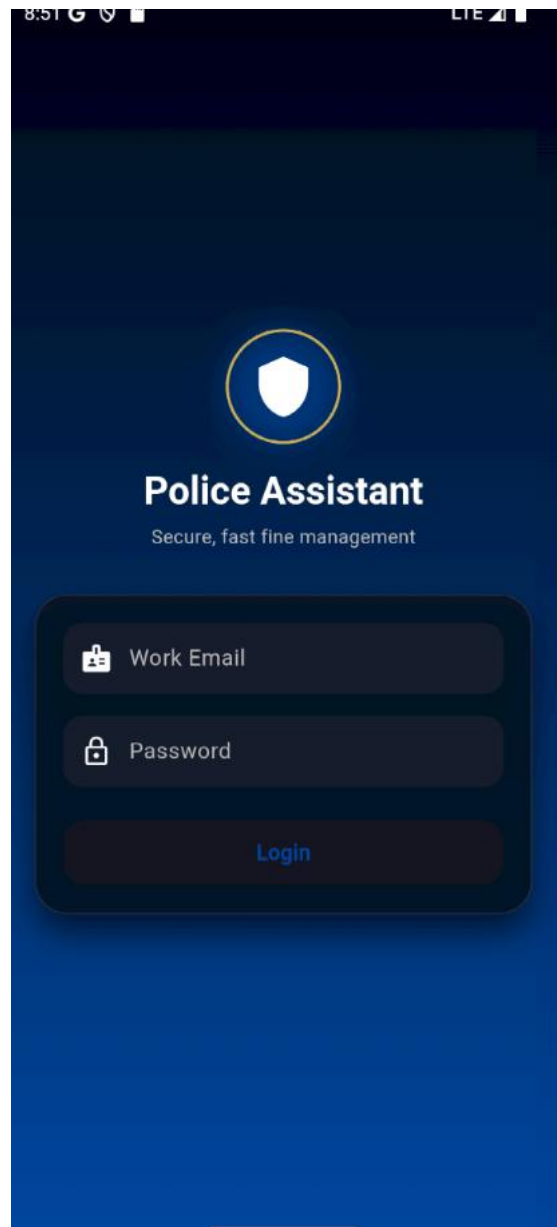
- **Visual Studio Code (VS Code)**

Visual Studio Code is a lightweight yet powerful source code editor that supports multiple programming languages and development tools. It provides features such as intelligent code completion, debugging, and extension support. In this project, VS Code is used to develop backend services, AI modules, and integration logic.

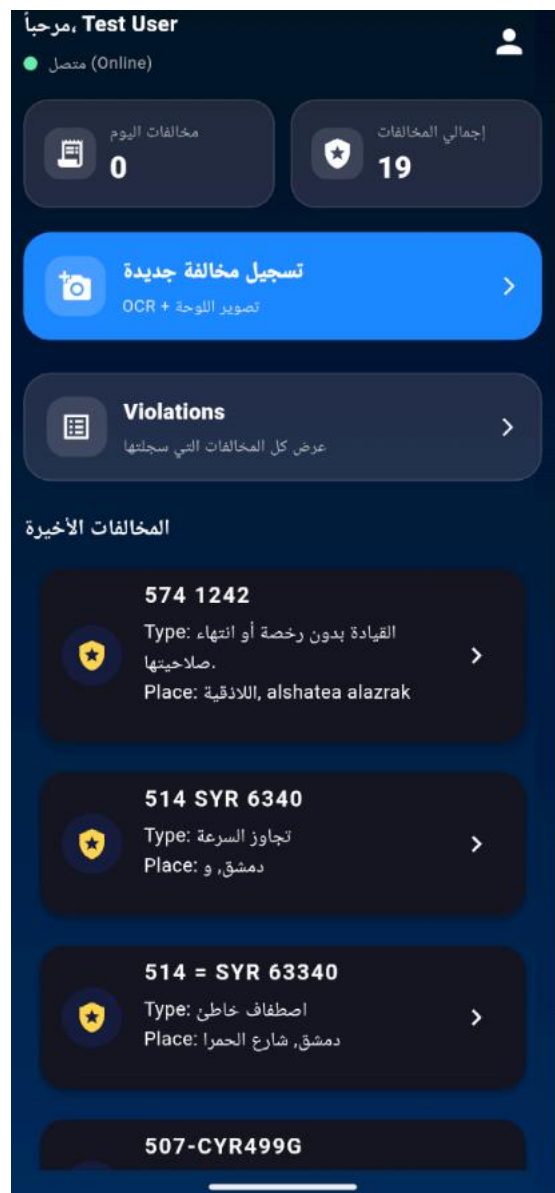
## System Interfaces:

### Police Officer interfaces:

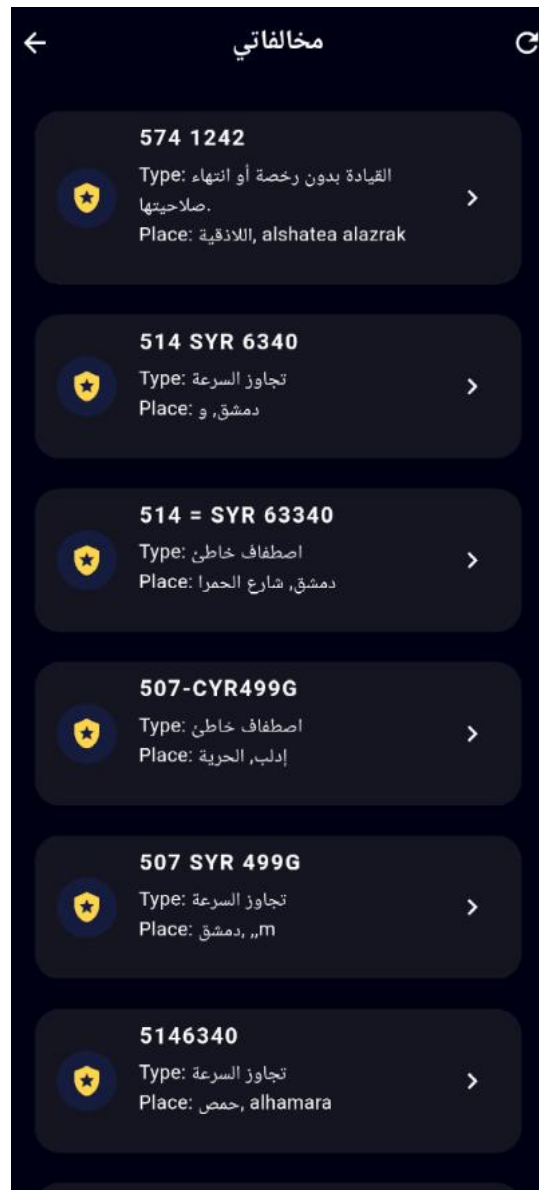
#### login interface:



## homepage interface:



## View Violation interface





## View Violation details interface

 **Violation Details**

 **574 1242**

**Vehicle Information**

 Plate  
574 1242

 Owner  
salam alrashed

**Location**

 City  
اللاذقية

 Street  
alshatea alazrak

 Landmark  
—

**Violation Details**

 Type  
القيادة بدون رخصة أو انتهاء صلاحيتها.

 Fine Amount  
100000

 Description  
—

 Date  
—

## Register violation page

9:34 G LTE

Register Violation

Add Car Image (OCR)

Voice input (STT)

Start Recording

Transcript (from STT)

Tip: After STT, fields may auto-fill if the backend returns structured data.

Plate Number

Owner Name (optional)

Car Model (optional)

Car Color (optional)

City  
دمشق

Street Name

Landmark (optional)

Register Violation

Tip: After STT, fields may auto-fill if the backend returns structured data.

Plate Number

Owner Name (optional)

Car Model (optional)

Car Color (optional)

City  
دمشق

Street Name

Landmark (optional)

Violation Type  
تجاوز السرعة

Description (optional)

Submit

## Profile settings interface:

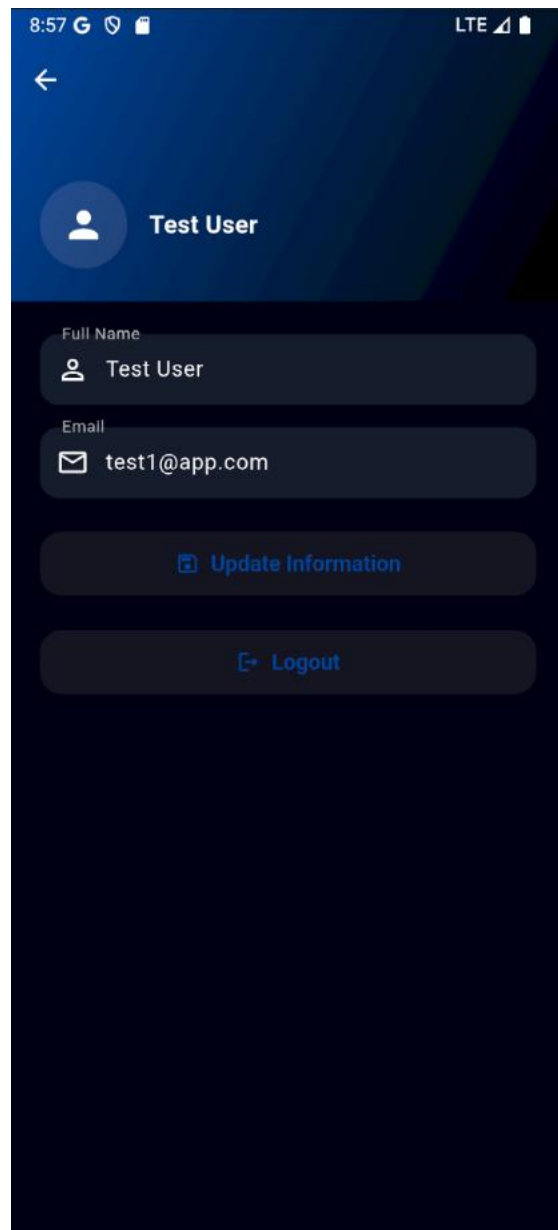
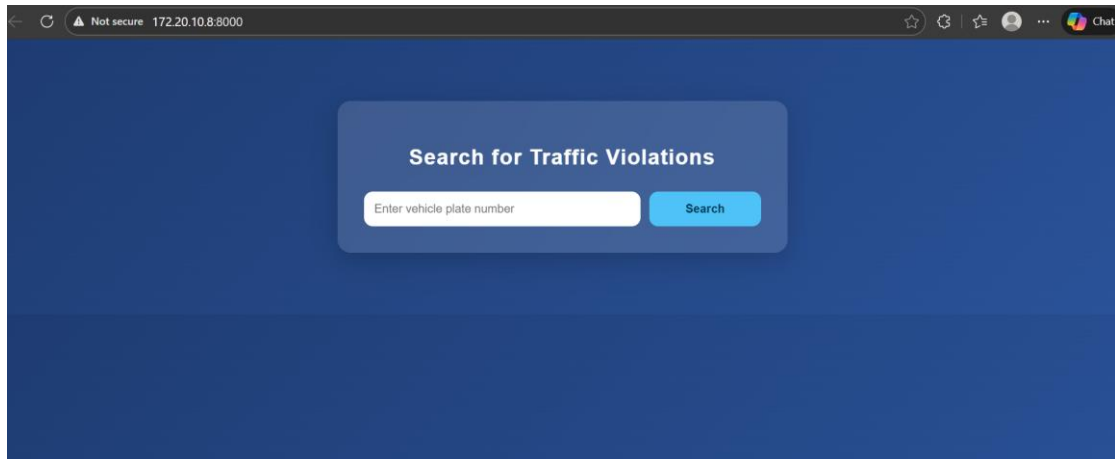


Figure 25 profile settings interface

## Citizen interfaces:

### Search for Violation

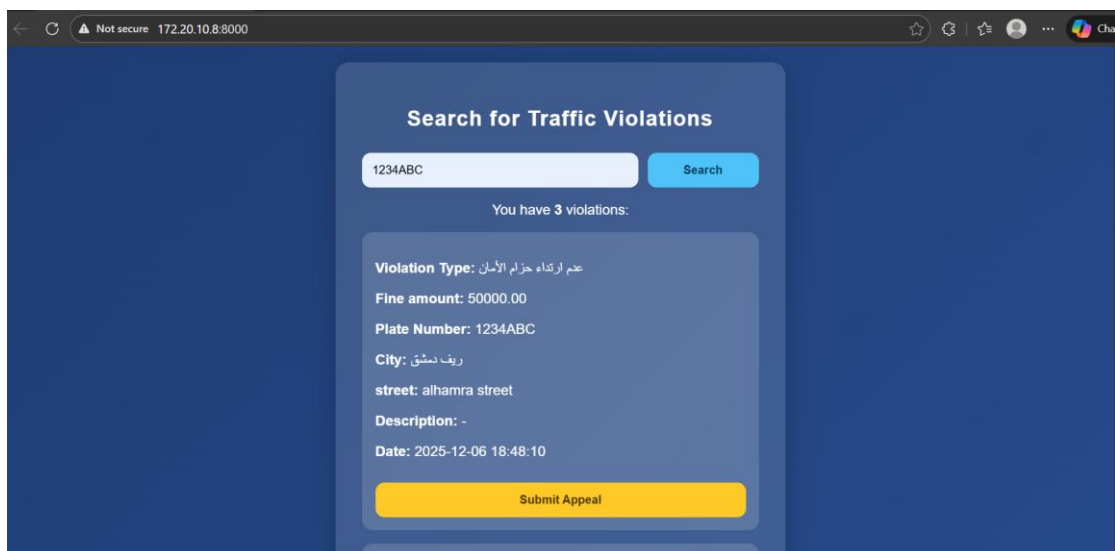


Search for Traffic Violations

Enter vehicle plate number

Search

### View violation list



Search for Traffic Violations

1234ABC

Search

You have 3 violations:

**Violation Type:** عم ارتداء حزام الأمان  
**Fine amount:** 50000.00  
**Plate Number:** 1234ABC  
**City:** ريف دمشق  
**street:** alhamra street  
**Description:** -  
**Date:** 2025-12-06 18:48:10

Submit Appeal

## Submit appeal

The screenshot shows a web browser window with the address bar displaying "Not secure 172.20.10.8:8000/citizen/appeal-form". The page has a dark blue background. In the center, there is a light blue rounded rectangle containing the following elements:

- A header "Violation Details" above a horizontal separator bar.
- A sub-header "Submit an Appeal".
- A label "Reason for Appeal:" followed by a text input field with the placeholder text "Describe why you are objecting".
- A blue button labeled "Send Appeal" at the bottom.

## View appeal states

The screenshot shows a web browser window with the address bar displaying "Not secure 172.20.10.8:8000". The page has a dark blue background. In the center, there is a light blue rounded rectangle containing the following information:

- Violation Type:** تجاوز السرعة
- Fine amount:** 150.00
- Plate Number:** 1234ABC
- City:** دمشق
- street:** شارع عدنان المالكي
- Description:** تجاوز السرعة
- Date:** 2025-02-10 14:30:00

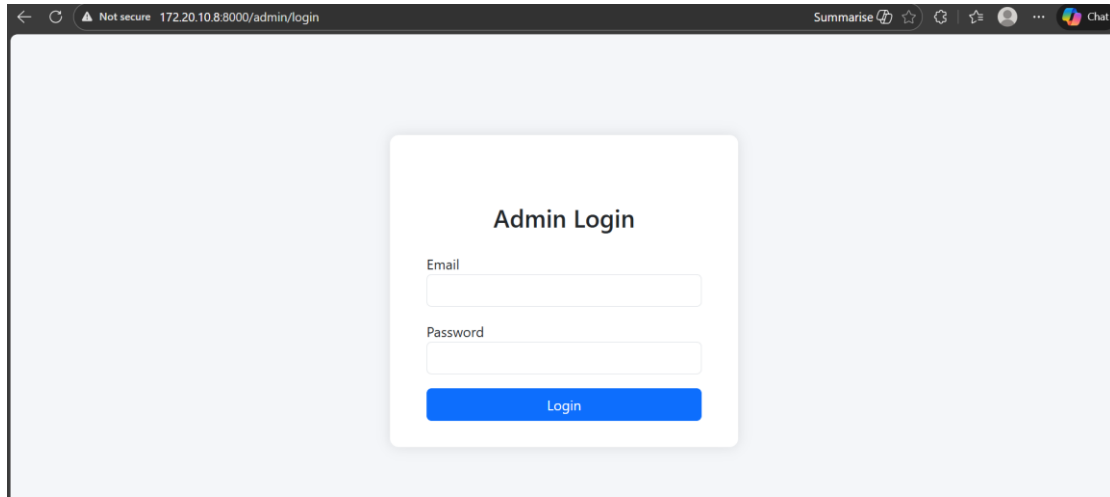
Below this information, there is a white box containing:

- Appeal Status:** pending
- Reason:** oliuytr
- Decision Note:** -
- Submitted:** 2025-12-06T13:39:38.000000Z

At the bottom of the light blue rectangle, the text "Violation Type: تجاوز السرعة" is visible.

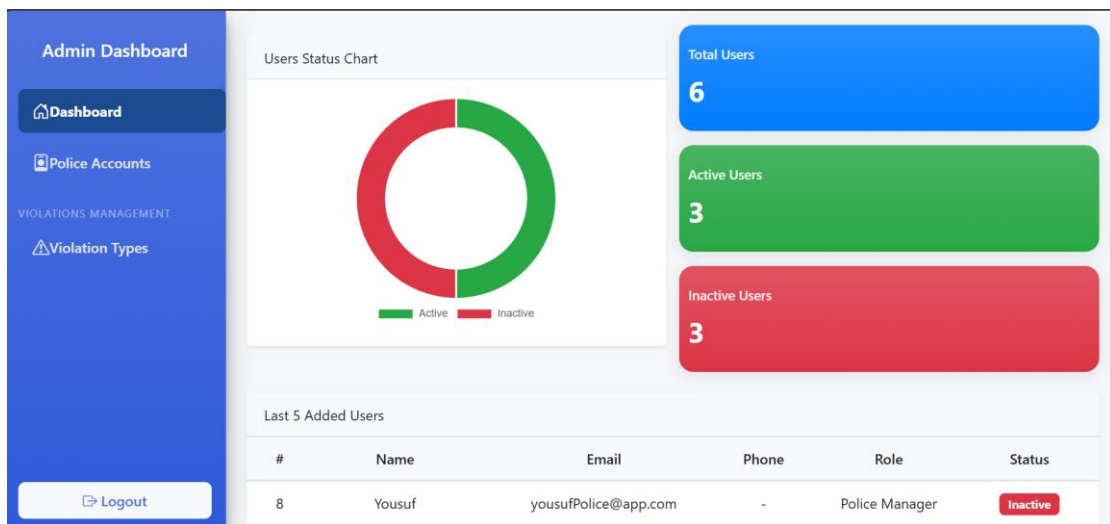
## Admin interface:

### Login interface :



A screenshot of a web browser showing the Admin Login interface. The browser's address bar displays '172.20.10.8:8000/admin/login'. The login form is centered on a light gray background and consists of a white box with the title 'Admin Login'. Inside the box, there are two input fields labeled 'Email' and 'Password', and a blue 'Login' button at the bottom.

### Dashboard interface :



The screenshot shows the Admin Dashboard with a sidebar containing 'Dashboard', 'Police Accounts', and 'Violation Types'. The main content area features a 'Last 5 Added Users' table and a summary card for 'Inactive Users'.

**Inactive Users Summary:**

| Inactive Users |
|----------------|
| 3              |

**Last 5 Added Users Table:**

| # | Name            | Email                | Phone | Role           | Status   |
|---|-----------------|----------------------|-------|----------------|----------|
| 8 | Yousuf          | yousufPolice@app.com | -     | Police Manager | Inactive |
| 7 | Sami Alhasan    | samiPolice@app.com   | -     | Police Manager | Inactive |
| 6 | Mohamed Alnajar | moPolice@app.com     | -     | Police Officer | Inactive |
| 4 | Test User       | test1@app.com        | -     | Police Officer | Active   |
| 1 | Test User       | test@example.com     | -     | Police Officer | Active   |

## Manage user account:

The screenshot shows the 'Police Accounts' management page. It includes a 'Create New Account' button and a table of users with 'Edit', 'Activate', 'Deactivate', and 'Delete' actions.

**Police Accounts Table:**

| # | Name            | Email                | Phone | Role           | Status   | Actions                  |
|---|-----------------|----------------------|-------|----------------|----------|--------------------------|
| 8 | Yousuf          | yousufPolice@app.com | —     | Police Manager | Inactive | Edit, Activate, Delete   |
| 7 | Sami Alhasan    | samiPolice@app.com   | —     | Police Manager | Inactive | Edit, Activate, Delete   |
| 6 | Mohamed Alnajar | moPolice@app.com     | —     | Police Officer | Inactive | Edit, Activate, Delete   |
| 4 | Test User       | test1@app.com        | —     | Police Officer | Active   | Edit, Deactivate, Delete |
| 1 | Test User       | test@example.com     | —     | Police Officer | Active   | Edit, Deactivate, Delete |

## Create user account:

The screenshot shows a web browser window with the URL `172.20.10.8:8000/admin/Users/create`. The browser's address bar also shows a security warning: "Not secure". The page features a blue sidebar on the left with the following navigation items: "Admin Dashboard", "Dashboard", "Police Accounts" (highlighted), "VIOLATIONS MANAGEMENT", and "Violation Types". At the bottom of the sidebar is a "Logout" button. The main content area has a blue header titled "Create New Police Officer". Below this header is a form with the following fields: "Name" (text input), "Email" (text input), "Phone (Optional)" (text input), "Password" (text input), "User Role" (dropdown menu with "Police Officer" selected), and "Account Status" (dropdown menu with "Active" selected). A blue "Create User" button is positioned at the bottom of the form.

## Edit user account:

The screenshot shows a web browser window with the URL `172.20.10.8:8000/admin/Users/edit/8`. The browser's address bar also shows a security warning: "Not secure". The page features a blue sidebar on the left with the following navigation items: "Admin Dashboard", "Dashboard", "Police Accounts" (highlighted), "VIOLATIONS MANAGEMENT", and "Violation Types". At the bottom of the sidebar is a "Logout" button. The main content area has a light blue header titled "Edit User". Below this header is a "Back to Users" button. The form contains the following fields: "Name" (text input with "yousuf" entered), "Email" (text input with "yousufPolice@app.com" entered), "Phone" (text input with "9 digits" entered), and "Account Status" (dropdown menu with "Active" selected). A blue "Save Changes" button is located at the bottom of the form.

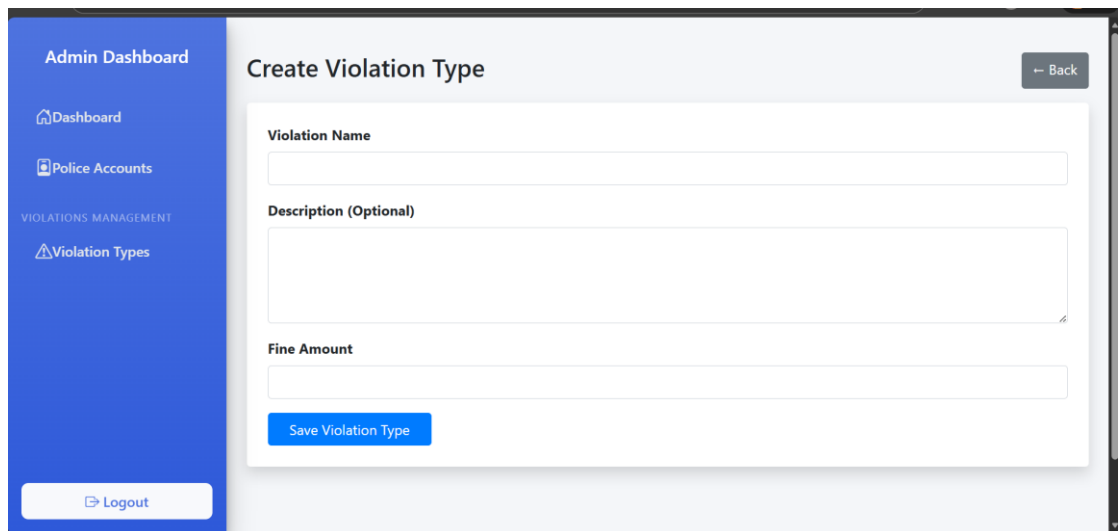


## View violation type



| # | Name                                  | Description | Fine Amount | Created At |
|---|---------------------------------------|-------------|-------------|------------|
| 1 | تجاوز السرعة                          |             | 150.00      | 2025-11-20 |
| 2 | اصطفاف خاطئ                           |             | 200000.00   | 2025-12-03 |
| 3 | القيادة بدون رخصة أو انتهاء صلاحيتها. |             | 100000.00   | 2025-12-06 |
| 4 | عدم ارتداء حزام الأمان                |             | 50000.00    | 2025-12-06 |
| 5 | مخالفة إشارات المرور                  |             | 150000.00   | 2025-12-06 |
| 6 | استخدام الهاتف أثناء القيادة          |             | 180000.00   | 2025-12-06 |
| 7 | test                                  | ...         | 150000.00   | 2025-12-06 |

## Create violation type



Admin Dashboard

Dashboard

Police Accounts

VIOLATIONS MANAGEMENT

Violation Types

Logout

### Create Violation Type

Back

Violation Name

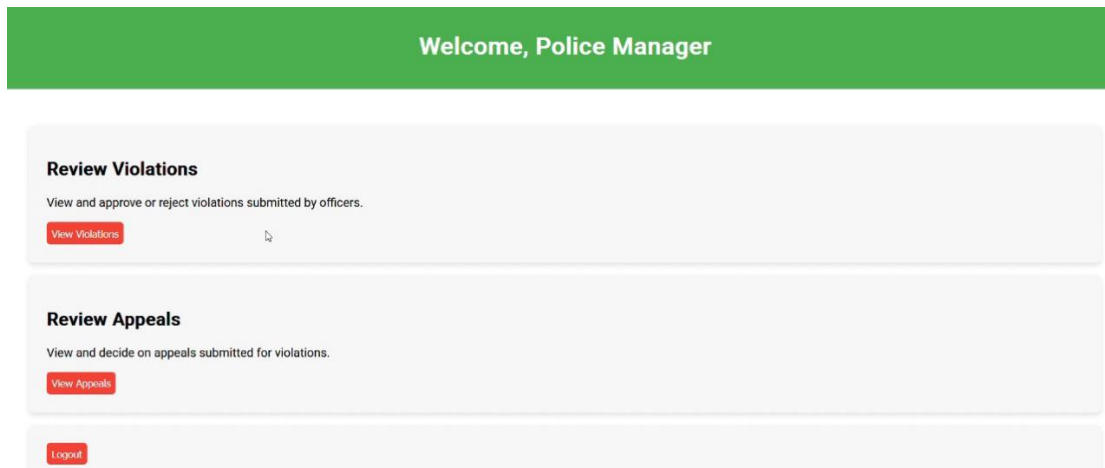
Description (Optional)

Fine Amount

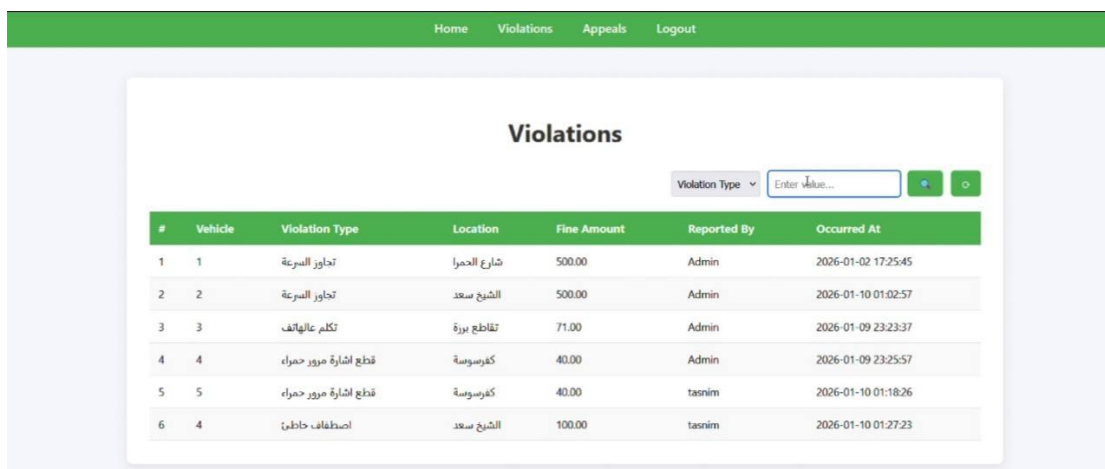
Save Violation Type

## Police manager interfaces:

### Police manager homepage



### View Violation interface



## Search by violation type

[Home](#) [Violations](#) [Appeals](#) [Logout](#)

### Violations

Violation Type

| # | Vehicle | Violation Type | Location     | Fine Amount | Reported By | Occurred At         |
|---|---------|----------------|--------------|-------------|-------------|---------------------|
| 1 | 1       | تجاوز السرعة   | شارع الحمراء | 500.00      | Admin       | 2026-01-02 17:25:45 |
| 2 | 2       | تجاوز السرعة   | الشيخ سعد    | 500.00      | Admin       | 2026-01-10 01:02:57 |

## Search by date

[Home](#) [Violations](#) [Appeals](#) [Logout](#)

### Violations

Occurred At

| # | Vehicle | Violation Type        | Location   | Fine Amount | Reported By | Occurred At         |
|---|---------|-----------------------|------------|-------------|-------------|---------------------|
| 1 | 3       | تكلم على الهاتف       | تقاطع برزة | 71.00       | Admin       | 2026-01-09 23:23:37 |
| 2 | 4       | قنطع إشارة مرور حمراء | كفرسوسة    | 40.00       | Admin       | 2026-01-09 23:25:57 |

## Search by reported by

[Home](#) [Violations](#) [Appeals](#) [Logout](#)

### Violations

Reported By

| # | Vehicle | Violation Type       | Location  | Fine Amount | Reported By | Occurred At         |
|---|---------|----------------------|-----------|-------------|-------------|---------------------|
| 1 | 5       | قطع إشارة مرور حمراء | كفرسوسة   | 40.00       | tasnim      | 2026-01-10 01:18:26 |
| 2 | 4       | اصطفاف خاطئ          | الشيخ سعد | 100.00      | tasnim      | 2026-01-10 01:27:23 |

## View appeals interface

[Home](#) [Violations](#) [Appeals](#) [Logout](#)

### Appeals List

| # | Violation ID | Status   | Reason                               | Created At          | Actions                             |
|---|--------------|----------|--------------------------------------|---------------------|-------------------------------------|
| 1 | 1            | Accepted | كنت واقف في الأوقات المسموحة بالوقوف | 2025-10-20 16:41:08 | <div>Accepted</div> <div>View</div> |
| 2 | 2            | accepted | كان لدي حالة طارئة                   | 2026-01-04 23:19:00 | <div>Pending</div> <div>View</div>  |
| 3 | 3            | accepted | حالة طارئة عندي                      | 2025-11-04 00:41:20 | <div>Pending</div> <div>View</div>  |

## Appeal details interface

[Home](#) [Violations](#) [Appeals](#) [Logout](#)

### Appeal Details

**Violation ID:** 1

**Status:** Accepted

**Reason:** كنت واقف في الاوقات المسوحة بالوقوف

**Decision Note:** تم قبول الاعتراض

**Created At:** 2025-10-20

Update Status: Accepted ▼

[Update Status](#)

[Back to Appeals List](#)

## Changing the appeal's status

[Home](#) [Violations](#) [Appeals](#) [Logout](#)

### Appeal Details

**Violation ID:** 1

**Status:** Accepted

**Reason:** كنت واقف في الاوقات المسوحة بالوقوف

**Decision Note:** تم قبول الاعتراض

**Created At:** 2025-10-20

Update Status: Accepted ▼

[Update Status](#) Pending

[Update Status](#) Accepted

[Update Status](#) Rejected

[Back to Appeals List](#)

## 4.updated test case :

Test Case Scenario: Sce-01: login/logout.

| Test case id | Test case title                                                    | Expected result                                                                                            | Tested data                                                                                     | Actual result                                | Pass/fail |
|--------------|--------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|----------------------------------------------|-----------|
| UC1-TC01     | Check successful login with valid credentials                      | Login is successful. The system grants access to the system main page. A user session is created.          | Email: <a href="mailto:tasnima.d2003@gmail.com">tasnima.d2003@gmail.com</a><br>password:simo123 | User redirected to main page. Access granted | pass      |
| UC1-TC02     | Check login fails with missing fields                              | The system displays an error message indicating a missing field and remains on the login page.             | Email:(empty)<br>password:simo123                                                               | Error message displayed                      | pass      |
| UC1-TC03     | Check login fails with invalid credentials (non-existent username) | The system displays an error message such as “Invalid username or password” and returns to the login form. | Email:wrongemail@gmail.com<br>password:simo123                                                  | Error message displayed                      | pass      |
| UC1-TC04     | Check login fails with invalid credentials (incorrect password)    | The system displays an error message such as “Invalid username or password” and returns to the login form. | Email:sara55@gmail.com<br>password:wrong pass                                                   | Error message displayed                      | pass      |
| UC1-TC05     | Check successful logout                                            | The user is logged out successfully. The session ends and the system returns to the login page.            | Logged-in user selects Logout                                                                   | Redirected to login page                     | pass      |

**Test Case Scenario: Sce-02: Account Management**

| Test case id | Test case title                                              | Expected result                                                                               | Tested data | Actual result                             | Pass / Fail |
|--------------|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------|-------------|-------------------------------------------|-------------|
| UC2-TC01     | Check view account profile                                   | The system displays the officer's account details correctly                                   | N/A         | Account details displayed correctly       | Pass        |
| UC2-TC02     | Check successful edit of account information with valid data | The system validates and saves the data successfully and displays a success message           | N/A         | Data saved and success message displayed  | Pass        |
| UC2-TC03     | Check edit account information with invalid data             | The system displays an error message and returns to the edit form                             | N/A         | Error message displayed                   | Pass        |
| UC2-TC04     | Check edit account information without making changes        | The system keeps the existing data unchanged and displays an appropriate confirmation message | N/A         | No changes applied, data remains the same | Pass        |

**Test Case Scenario: Sce-03: Generate violation:**

| Test case id | Test case title                                            | Expected result                                                                                                                    | Tested data                                                                          | Actual result                              | Pass / Fail |
|--------------|------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------|-------------|
| UC3-TC01     | Check successful generation of violation using audio input | The system converts the recorded audio to text, extracts vehicle plate number, generates the violation, and saves it successfully. | Clear audio recording, clear vehicle image, selected violation type(speed exceeding) | Violation generated and saved successfully | Pass        |
| UC3-TC02     | Check audio recording failure during violation generation  | The system detects recording failure and requests the officer to re-record the audio.                                              | No audio captured                                                                    | Re-recording request displayed             | Pass        |
| UC3-TC03     | Check text conversion failure after audio recording        | The system allows the officer to manually enter the violation text.                                                                | Audio recorded but conversion failed                                                 | Manual text entry option displayed         | Pass        |
| UC3-TC04     | Check license plate extraction failure                     | The system requests the officer to manually enter the license plate number.                                                        | Blurry vehicle image                                                                 | Manual plate entry requested               | Pass        |
| UC3-TC05     | Check review of generated violation before confirmation    | The system displays the generated violation details for review.                                                                    | Generated violation data                                                             | Violation details displayed                | Pass        |



**Test Case Scenario: Sce-04: Add violation:**

| Test case id | Test case title                                          | Expected result                                                                                      | Tested data                                                          | Actual result                                     | Pass / Fail |
|--------------|----------------------------------------------------------|------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|---------------------------------------------------|-------------|
| UC4-TC01     | Check successful addition of a violation with valid data | The system validates the data, generates the violation report, and saves the violation successfully. | Valid violation details (correct format, all required fields filled) | Violation report generated and saved successfully | Pass        |
| UC4-TC02     | Check add violation with missing required fields         | The system displays an error message and returns to the violation data entry form.                   | Missing required fields(audio/plate number/violation type)           | Error message displayed and form returned         | Pass        |
| UC4-TC03     | Check add violation with invalid data                    | The system displays an error message and allows the officer to correct the entered data.             | Invalid data format (audio/plate number/violation type)              | Validation error displayed                        | Pass        |

**Test Case Scenario: Sce-05: view violation:**

| Test case id | Test case title                                    | Expected result                                                                                         | Tested data                                                         | Actual result                           | Pass / Fail |
|--------------|----------------------------------------------------|---------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|-----------------------------------------|-------------|
| UC5-TC01     | Check viewing list of existing violations          | The system displays the list of existing violations successfully.                                       | Logged-in police officer account with at least one saved violation  | List of existing violations displayed   | Pass        |
| UC5-TC02     | Check viewing violation details                    | The system displays the detailed information of the selected violation.                                 | Selected violation record from violations list                      | Violation details displayed correctly   | Pass        |
| UC5-TC03     | Check searching violations by date                 | The system displays the violations that match the selected date and shows the violation details.        | Search date that matches one or more stored violations( 30-10-2025) | Matching violations displayed           | Pass        |
| UC5-TC04     | Check searching violations by license plate number | The system displays the violations that match the entered plate number and shows the violation details. | License plate number associated with existing violation( 1223240)   | Matching violations displayed           | Pass        |
| UC5-TC05     | Check viewing violations when none exist           | The system displays the message “No violations found”.                                                  | Logged-in police officer account with zero violations               | “No violations found” message displayed | Pass        |

**Test Case Scenario: Sce-06: Create Police Account:**

| Test case id | Test case title                                               | Expected result                                                                                                            | Tested data                                                                                                                | Actual result                                                  | Pass / Fail |
|--------------|---------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|-------------|
| UC6-TC01     | Check successful creation of a police account with valid data | The system validates the data, saves the new police account, and displays a successful creation confirmation.              | Valid and complete police officer details (unique username: Maryam<br>valid email: maryam@example.com, valid officer ID:1) | Police account created successfully and confirmation displayed | Pass        |
| UC6-TC02     | Check police account creation with missing required data      | The system displays an error message and returns to the registration form to correct the data.                             | One or more required fields left empty ( example empty name)                                                               | Error message displayed and registration form shown again      | Pass        |
| UC6-TC03     | Check police account creation with duplicate data             | The system displays an error message indicating duplicate data and allows the administrator to re-enter valid information. | Existing username or officer ID already stored in the system                                                               | Duplicate data error message displayed                         | Pass        |
| UC6-TC04     | Check police account creation with invalid data               | The system displays an error message indicating invalid data and returns to the registration form for correction           | Invalid email format: sara333@example.com<br>and/or invalid officer ID format:10                                           | Validation error message displayed                             | Pass        |

**Test Case Scenario: Sce-07: edit Police Account:**

| Test case id | Test case title                                             | Expected result                                                                                                | Tested data                                                    | Actual result                                                          | Pass / Fail |
|--------------|-------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|------------------------------------------------------------------------|-------------|
| UC7-TC01     | Check successful editing of police account with valid data. | The system validates the data and updates the police account successfully. A confirmation message is displayed | Existing police account with valid updated details(name/email) | Police account updated successfully and confirmation message displayed | Pass        |
| UC7-TC02     | Check editing police account with missing required fields   | The system displays an error message indicating missing required fields and requests correction                | One or more required fields left empty during editing          | Error message displayed requesting correction                          | Pass        |
| UC7-TC03     | Check editing police account with invalid data format       | The system rejects the input and displays a validation error message.                                          | Invalid data format (e.g., incorrect phone number format)      | Validation error message displayed                                     | Pass        |

**Test Case Scenario: Sce-o8: Activate / Deactivate Police Account:**

| Test case id | Test case title                                       | Expected result                                                                                   | Tested data                                                         | Actual result                                             | Pass / Fail |
|--------------|-------------------------------------------------------|---------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|-----------------------------------------------------------|-------------|
| UC8-TC01     | Check successful activation of a police account       | The system activates the police account successfully and updates the account status to Active     | Deactivated police account selected                                 | Police account activated and status updated to Active     | Pass        |
| UC8-TC02     | Check successful deactivation of a police account     | The system deactivates the police account successfully and updates the account status to Inactive | Active police account selected                                      | Police account deactivated and status updated to Inactive | Pass        |
| UC8-TC03     | Check status update reflected in police accounts list | The updated account status is correctly reflected in the police accounts list                     | If the police account is active update to inactive and the opposite | Updated status displayed correctly in accounts list       | Pass        |

**Test Case Scenario: Sce-09: Review Violations:**

| Test case id | Test case title                                            | Expected result                                                  | Tested data                                       | Actual result                          | Pass / Fail |
|--------------|------------------------------------------------------------|------------------------------------------------------------------|---------------------------------------------------|----------------------------------------|-------------|
| UC9-TC01     | Check successful display of violations list                | The system displays the list of violations successfully.         | Logged-in Police Manager with existing violations | Violations list displayed successfully | Pass        |
| UC9-TC02     | Check successful search and review of a specific violation | The system displays the selected violation details successfully. | Valid violation selected from the list            | Violation details displayed correctly  | Pass        |
| UC9-TC03     | Check when no matching violation is found                  | The system displays a message "No                                | Invalid or non-existing search criteria           | "No matching violations found"         | Pass        |

|  |  |                             |  |                   |  |
|--|--|-----------------------------|--|-------------------|--|
|  |  | matching violations found". |  | message displayed |  |
|--|--|-----------------------------|--|-------------------|--|

**Test Case Scenario: Sce-10: Review appeals:**

| Test case id | Test case title                                      | Expected result                                                                         | Tested data                                                                | Actual result                       | Pass / Fail |
|--------------|------------------------------------------------------|-----------------------------------------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------|-------------|
| UC10-TC01    | Check display of appeals list                        | The system displays the list of appeals successfully.                                   | Police Manager logged in, system contains appeals with different statuses  | Appeals list displayed successfully | Pass        |
| UC10-TC02    | Check display of appeal details                      | The system displays the appeal details successfully.                                    | Existing appeal selected from the appeals list                             | Appeal details displayed correctly  | Pass        |
| UC10-TC03    | Check successful acceptance of an appeal             | The system records the decision successfully and updates the appeal status to Accepted. | Appeal with status <b>Pending</b> , decision selected = Accept             | Appeal status updated to Accepted   | Pass        |
| UC10-TC04    | Check successful rejection of an appeal              | The system records the decision successfully and updates the appeal status to Rejected. | Appeal with status <b>Pending</b> , decision selected = Reject             | Appeal status updated to Rejected   | Pass        |
| UC10-TC05    | Check system behavior when decision submission fails | The system displays the message "Error submitting decision".                            | Appeal with status <b>Pending</b> , decision submitted during system error | Error message displayed             | Pass        |

**Test Case Scenario: Sce-11: view violation:**

| Test case id | Test case title                                                  | Expected result                                                                                    | Tested data                                                       | Actual result                                         | Pass / Fail |
|--------------|------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|-------------------------------------------------------|-------------|
| UC11-TC01    | Check successful viewing of violations with valid vehicle number | The system displays the violations list and shows the selected violation details successfully      | Valid vehicle number(1998065) associated with existing violations | Violations list displayed and violation details shown | Pass        |
| UC11-TC02    | Check system behavior with invalid vehicle number                | The system displays an error message indicating invalid vehicle number and no violations are shown | Invalid or incorrectly formatted vehicle number(000000)           | Error message displayed and no violations shown       | Pass        |
| UC11-TC03    | Check system behavior when no violations exist                   | The system displays the message "No violations found"                                              | Valid vehicle number with no associated violations                | "No violations found" message displayed               | Pass        |
| UC11-TC04    | Check display of violations list                                 | The system displays a list of all violations related to the entered vehicle number                 | Valid vehicle number with multiple associated violations          | Violations list displayed successfully                | Pass        |
| UC11-TC05    | Check display of violation details                               | The system displays full details of the selected violation                                         | Valid vehicle number and selected violation from list             | Violation details displayed correctly                 | Pass        |

**Test Case Scenario: Sce-12: submit appeal:**

| Test case id | Test case title                                                              | Expected result                                                                                   | Tested data                                                      | Actual result                                                | Pass / Fail |
|--------------|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|------------------------------------------------------------------|--------------------------------------------------------------|-------------|
| UC12-TC01    | Check successful submission of an appeal with valid data                     | The system validates the data, saves the appeal successfully, and displays a confirmation message | Valid violation selected, valid appeal reason, valid notes       | Appeal saved successfully and confirmation message displayed | Pass        |
| UC12-TC02    | Check submission of appeal with incomplete data                              | The system displays the message "Incomplete data" and the appeal is not saved                     | Violation selected, missing required fields (reason or notes)    | "Incomplete data" message displayed and appeal not saved     | Pass        |
| UC12-TC03    | Check system behavior when a technical error occurs during appeal submission | The system displays the message "Unable to submit appeal currently" and the appeal is not saved   | Valid violation selected, valid appeal data, system/server error | Error message displayed and appeal not saved                 | Pass        |
| UC12-TC04    | Check display of appeal submission form                                      | The system displays the appeal submission form correctly                                          | Valid violation selected                                         | Appeal submission form displayed correctly                   | Pass        |
| UC12-TC05    | Check that appeal is linked to the correct violation                         | The system links the submitted appeal to the selected violation correctly                         | Valid violation selected and appeal submitted                    | Appeal linked to the correct violation                       | Pass        |



## RTM Last Version:

| Req-ID   | Use Case | Analysis | System Design             | Detailed Design | Coding | Test Case(s)       |
|----------|----------|----------|---------------------------|-----------------|--------|--------------------|
| RE-FR-01 | UC-1     | AM-AN    | User Management Service   | —               | Code   | UC1-TC01, UC1-TC02 |
| RE-FR-02 | UC-1     | AM-AN    | User Management Service   | —               | Code   | UC1-TC03           |
| RE-FR-03 | UC-2     | AM-AN    | User Management Service   | —               | Code   | UC2-TC01, UC2-TC02 |
| RE-FR-04 | UC-3     | AM-AN    | Violation + AI Services   | —               | Code   | UC3-TC01           |
| RE-FR-05 | UC-4     | AM-AN    | Violation Management      | —               | Code   | UC4-TC01, UC4-TC02 |
| RE-FR-06 | UC-3     | AM-AN    | Plate Recognition Service | —               | Code   | UC3-TC02, UC3-TC03 |
| RE-FR-07 | UC-3     | AM-AN    | Violation Management      | —               | Code   | UC3-TC04           |
| RE-FR-08 | UC-3     | AM-AN    | Violation Management      | —               | Code   | UC3-TC05           |
| RE-FR-09 | UC-5     | BM-AN    | Violation Management      | —               | Code   | UC5-TC01           |
| RE-FR-10 | UC-5     | BM-AN    | Violation Management      | —               | Code   | UC5-TC02           |
| RE-FR-11 | UC-11    | BM-AN    | Violation Management      | —               | Code   | UC11-TC01          |
| RE-FR-12 | UC-12    | BM-AN    | Appeal Management         | —               | Code   | UC12-TC01          |
| RE-FR-13 | UC-12    | BM-AN    | Appeal Management         | —               | Code   | UC12-TC02          |
| RE-FR-14 | UC-1     | AM-AN    | User Management Service   | —               | Code   | UC1-TC01           |

|          |       |        |                           |   |      |           |
|----------|-------|--------|---------------------------|---|------|-----------|
| RE-FR-15 | UC-1  | AM-AN  | User Management Service   | — | Code | UC1-TC03  |
| RE-FR-16 | UC-2  | AM-AN  | User Management Service   | — | Code | UC2-TC01  |
| RE-FR-17 | UC-6  | BM-AN  | User Management Service   | — | Code | UC6-TC01  |
| RE-FR-18 | UC-7  | BM-AN  | User Management Service   | — | Code | UC7-TC01  |
| RE-FR-19 | UC-7  | BM-AN  | User Management Service   | — | Code | UC7-TC02  |
| RE-FR-20 | UC-8  | BM-AN  | User Management Service   | — | Code | UC8-TC01  |
| RE-FR-21 | UC-6  | BM-AN  | User Management Service   | — | Code | UC6-TC02  |
| RE-FR-22 | UC-10 | DM-AN  | Appeal Management         | — | Code | UC10-TC01 |
| RE-FR-23 | UC-10 | DM-AN  | Appeal Management         | — | Code | UC10-TC02 |
| RE-FR-24 | UC-10 | DM-AN  | Appeal Management         | — | Code | UC10-TC03 |
| RE-FR-25 | UC-3  | FGM-AN | AI – Speech to Text       | — | Code | UC3-TC06  |
| RE-FR-26 | UC-3  | FGM-AN | AI – Plate Recognition    | — | Code | UC3-TC07  |
| RE-FR-27 | UC-3  | FGM-AN | AI – Violation Generation | — | Code | UC3-TC08  |
| RE-FR-28 | UC-3  | FGM-AN | AI – Classification       | — | Code | UC3-TC09  |

## **Chapter 8 Conclusion**

## Introduction

As a result of this project, we have developed a fully functional **Traffic Police Assistant System** designed to support traffic police officers in efficiently recording, managing, and reviewing traffic violations. The system aims to replace traditional manual processes with a smart, technology-driven solution that improves accuracy, speed, and reliability in field operations.

The system enables officers to capture vehicle images and automatically extract critical information such as license plate numbers, vehicle model, and color using **Optical Character Recognition (OCR)** technologies. In addition, officers can record verbal descriptions of violations through **Speech-to-Text (STT)** functionality, allowing violation details to be transcribed automatically and used to populate violation records. This significantly reduces manual data entry and enhances usability in real-world traffic scenarios.

To ensure scalability, maintainability, and clean separation of concerns, the backend was developed following modern software engineering principles and a service-oriented architecture. The system integrates a RESTful API built with **Laravel**, a relational database for structured data management, and **RabbitMQ** to handle asynchronous AI tasks such as OCR and STT processing. This architecture allows the system to efficiently manage heavy workloads and supports future extensions.

Throughout the development of this project, our main objectives were to:

Improve the efficiency and accuracy of traffic violation recording.

Reduce manual effort for traffic police officers in the field.

Provide a secure, user-friendly, and reliable system.

Ensure scalability and extensibility through clean, modular design and modern backend practices.

### References

[Laravel Documentation \(Latest Version\)](#)

[PHP Official Documentation](#)

[Flutter Documentation](#)

[RabbitMQ Documentation](#)

[RESTful API Design Best Practices](#)

[GitHub Documentation](#)

### Appendices

## GitHub Repositories

### Backend:

[https://github.com/almasri-maryam/Triffic\\_Police\\_assistant.git](https://github.com/almasri-maryam/Triffic_Police_assistant.git)

### Frontend (Flutter App):

[https://github.com/simo75-t/grad\\_project.git](https://github.com/simo75-t/grad_project.git)